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Predicting the Goal-Plane Entry of a Penalty Kick from Two Early Ball Positions

Abstract

A penalty kick is decided before the ball has travelled three metres, and the goalkeeper's side is committed about 0.1 s before the kick. We ask whether the goal-plane crossing point can be recovered from two measured positions, at 0.05 s and 0.10 s after the kick. A six-parameter transverse-force model, comprising a launch velocity, a drag coefficient and two transverse coefficients, is fitted to the six coordinates by our own damped Gauss-Newton routine at an assumed noise of 5 mm. The fitted trajectory crosses the plane at $y = 2.785$ m, $z = 1.945$ m, $t = 0.399$ s, 0.87 m inside the post and 0.50 m below the crossbar, with 90 per cent intervals $y \in [2.455, 3.121]$ m and $z \in [1.617, 2.276]$ m; the whole ball passes inside the frame with probability 0.971. The four complete models place the point between 2.549 and 2.785 m, and the model envelope of calibrated intervals is $y \in [2.125, 3.121]$ m.

Imposing extreme assumptions widens this to a scenario envelope, $y \in [2.259, 3.935]$ m and $z \in [1.169, 2.698]$ m, which is a stress range and not a confidence interval. Coverage is tested on the estimator itself, with truths from the same law and from a law containing an extra vertical force. The method is also a design statement: the two epochs of the problem give $\sigma_y = 211$ mm, while fixes at 0.11 s and 0.31 s give 12 mm, a factor of 18. The prediction is ready 0.11 s after the kick, 135 ms before the goalkeeper can resolve height and 0.29 s before the ball arrives.

Keywords: penalty kick, goal-plane prediction, Magnus force, inverse problem, Bayesian uncertainty quantification, identifiability

1 Introduction

A penalty kick is decided in about half a second. The ball leaves the foot at about 33 m/s and crosses the goal plane 0.4 s to 0.5 s later, whereas the goalkeeper's dive direction is differentiated about 115 ms *before* contact and the height of the shot only 245 ms *after* it [1]. A forecast issued between 0.1 s and 0.2 s after contact therefore arrives while the height of the shot is still being resolved, though the side of the dive is already fixed.

The measurement layer exists. Goal-line technology must locate the ball to ± 1.5 cm within one second [2], and Hawk-Eye's seven cameras per goal give a three-dimensional fix from any two of them [3]; broadcast tracking is certified at 100 Hz but only for two-dimensional ball position and speed [4]. None measures spin, and none predicts. The two fixes supplied with this problem are exactly such data; the missing piece is the inversion to a goal-plane forecast.

The inversion is elementary in form: two positions with a known origin give the velocity by a first difference and the acceleration by a second. With independent position noise $\sigma_{\mathrm{pos}} = 5$ mm—half of the last printed digit, as A6 explains—the velocity is fixed to $\sqrt{2}\,\sigma_{\mathrm{pos}}/\Delta t \approx 0.14$ m/s, under half a per cent of the launch speed, whereas the acceleration is fixed only to $\sqrt{6}\,\sigma_{\mathrm{pos}}/\Delta t^2 \approx 4.9\,\mathrm{m/s^2}$, about 30 per cent of the $16\,\mathrm{m/s^2}$ lateral component the data imply. The launch state is measured; the force is not. Six observations absorb a constant acceleration exactly, $\boldsymbol{a} = (-16.000, -16.000, -12.000)\ \mathrm{m/s^2}$, but the same six numbers underdetermine any richer force law. A seven-parameter model has a data Jacobian of rank six, with three singular values near 300 and three near 1.8; the data determine about 5.3 parameters, and the coefficient along the instantaneous airflow is shrunk by only 8.7 per cent. Moving that coefficient over $\pm 0.075\,\mathrm{m^{-1}}$, a range the data cannot distinguish, moves the predicted entry from $y = 2.56$ m to 2.78 m and from $z = 2.12$ m to 1.85 m. Linear propagation of parameter uncertainty therefore does not describe the prediction error: the error is dominated by the choice of force law, not by the width of a parameter interval [5, 6].

Only a force perpendicular to the velocity can bend the shot. Drag is antiparallel to the velocity and gravity is vertical, so a drag-only model preserves v_y/v_x exactly and returns $y = x_g v_{0y}/v_{0x}$ for every drag coefficient: the gravity-only and gravity-plus-drag fits give $y = 3.2170$ m and 3.2172 m. The bending force is the Magnus force, with measured lift coefficient 0.21–0.29 over spin parameters 0.18–0.34 [7]; a non-spinning ball also feels a lateral force of comparable size from wake asymmetry [8]; spin decays by only 5 % to 15 % over 18 m [9]; and spin is not observable from positions alone [10]. Past the drag crisis, between $\mathrm{Re} \approx 9 \times 10^4$ and 1.3×10^5, C_D is nearly flat at 0.17–0.22 [11–13], so the flight is well described by a constant C_D.

Other sports have inverted the same physics with more data. Drag has been recovered from whole projectile trajectories [14] and from trajectory curvature [15]; 99 tracked baseball pitches yield C_D, C_L and the spin vector [16]; spin needs dedicated marker or multi-view measurement even then [17]; three-dimensional reconstruction of football flight from broadcast video is routine [18, 19]; and the penalty has been modelled forward [20]. Those studies either measure the aerodynamic parameters or average over a whole flight; propagating uncertain aerodynamic parameters to an impact point is a separate, studied problem [21, 22], and the present case is its two-observation limit.

Applied to the given fixes, a six-parameter transverse-force fit of the launch velocity, the drag coefficient and two transverse coefficients places the ball centre at

$(y, z) = (2.785, 1.945)$ m at $t = 0.399$ s, and the whole ball passes inside the frame with probability 0.971. The value is conditional on the force law: removing the transverse force moves the entry to $y = 3.22$ m, a literature-prior fit gives 3.03 ± 0.28 m, and a constant-acceleration extrapolation gives 2.50 ± 0.60 m with $P = 0.73$. We therefore report three separate ranges rather than one. The model envelope unions the 90 per cent intervals of the complete models, $y \in [2.125, 3.121]$ m and $z \in [1.617, 2.276]$ m. The scenario envelope unions the 95 per cent intervals of twenty-five single-factor variants, four joint-stress cases and an independent replication, giving $y \in [2.259, 3.935]$ m and $z \in [1.169, 2.698]$ m; it is a stress range, not a confidence or credible interval.

2 Problem Restatement and Notation

The penalty mark is the origin of a Cartesian frame whose x-axis points at the centre of the goal and whose z-axis is vertical. The ball has radius $R = 0.110$ m, and at the kick its centre is at $\boldsymbol{r}_0 = (0, 0, 0.11)$ m, one radius above the ground. The goal plane is $x = 11.0$ m; the opening between the posts is $|y| \le 3.66$ m and the space under the crossbar is $0 \le z \le 2.44$ m. The ball centre is observed twice, at $t_1 = 0.05$ s and $t_2 = 0.10$ s:

$$\boldsymbol{r}_1 = (1.50,\ 0.45,\ 0.44), \qquad \boldsymbol{r}_2 = (2.96,\ 0.86,\ 0.74), \tag{1}$$

with all coordinates in metres. These numbers and the geometry of Figure 1 are the entire input; Table 1 collects them. All times are measured from the kick, and every quantity is a Cartesian component; SI units are used throughout.

Each reported coordinate is taken to carry independent Gaussian noise of standard deviation $\sigma_{\text{pos}} = 0.005$ m, isotropic in the three directions. The coordinates are printed to 0.01 m, so half of the last printed digit is the only defensible scale; the number describes the precision of the printed input, not a camera calibration, and it is not the ± 1.5 cm accuracy required of goal-line technology [2], which bounds a certified installation rather than this problem's data. The choice and its consequences are examined in A6 of Section 3. The fixes are samples of the ball centre with no common offset (A7), and the kick position is exact. No direction is assumed for the launch: the initial velocity $\boldsymbol{v}_0$ is unknown.

Vectors are bold. The unknown initial velocity is $\boldsymbol{v}_0 = (v_{0x}, v_{0y}, v_{0z})$; the constant acceleration used by the control estimator is $\boldsymbol{a}$; the ball's angular velocity is $\boldsymbol{\omega} = (\omega_x, \omega_y, \omega_z)$ with magnitude $\omega = |\boldsymbol{\omega}|$. The flight models are written in terms of the drag coefficient C_D, the lift coefficient $C_L = C_{L0} S_p$ with $S_p = \omega R / |\boldsymbol{v}|$ the spin parameter, the spin-axis unit vector $\hat{\boldsymbol{\omega}}$, and the spin-decay time τ, with $\omega(t) = \omega_0 \mathrm{e}^{-t/\tau}$; the remaining constants are the air density $\rho = 1.20\,\mathrm{kg/m^3}$, the ball mass $m = 0.430$ kg, the cross-sectional area $A = \pi R^2$, and gravity $g = 9.81\,\mathrm{m/s^2}$. The transverse-force fit that supplies the reported answer replaces the three spin components by two coefficients $\boldsymbol{\beta} = (\beta_1, \beta_2)$, in $\mathrm{m^{-1}}$, of the transverse acceleration, so that its parameter vector is $\boldsymbol{\theta} = (\boldsymbol{v}_0, C_D, \boldsymbol{\beta})$; the literature-prior control uses $(\boldsymbol{v}_0, C_D, \boldsymbol{\omega})$, and the constant-acceleration control uses $(\boldsymbol{v}_0, \boldsymbol{a})$. The crossing time $t_\star$ is defined by $x(t_\star) = 11.0$ m.

The task is to turn these inputs into three linked outputs: a point estimate $(\hat{y}, \hat{z})$ of the ball-centre crossing and the crossing time $\hat{t}_\star$; a predictive distribution for (y, z) at the plane, reported as standard deviations and as 90 and 95 per cent intervals; and the probability that the ball scores. The point estimate is the most likely entry point, i.e. the mode of the predictive distribution; where that distribution is nearly Gaussian the mode and the mean nearly coincide,

and the mean may be quoted. Scoring is interpreted in the strict sense of the Laws: the whole ball must pass inside the frame, so a clean goal requires $|y| \le 3.66 - R = 3.55\,\mathrm{m}$ and $R \le z \le 2.44 - R = 2.33\,\mathrm{m}$ [23]. All probabilities quoted in this paper use this convention; a convention based on the ball centre alone would enlarge the effective target by one radius on each side. Three estimators are compared: the transverse-force fit, which supplies the reported answer; the constant-acceleration estimator; and the literature-prior Bayesian fit. The reported uncertainty is the model envelope of Section 3, a range of plausible answers under different model forms and priors rather than one estimator's interval.

Table 1: Given quantities and goal geometry. Positions are ball-centre coordinates in metres.

Quantity	Symbol	Value
Kick instant	t_0	0
Ball centre at kick	$\boldsymbol{r}_0$	$(0, 0, 0.11)$ m
First fix	$\boldsymbol{r}_1$	$(1.50, 0.45, 0.44)$ m at $t_1 = 0.05\,\mathrm{s}$
Second fix	$\boldsymbol{r}_2$	$(2.96, 0.86, 0.74)$ m at $t_2 = 0.10\,\mathrm{s}$
Position noise per coordinate	σ_{pos}	0.005 m (assumed)
Goal plane	x	11.0 m
Goal opening	$\lvert y\rvert$, z	$\le 3.66\,\mathrm{m}$, 0–2.44 m
Ball radius	R	0.110 m

3 Assumptions and Their Consequences

Every answer in this paper is conditional on the eight assumptions below. Each is stated, justified against the literature or the problem data, and followed by the change in the answer if it fails. Where the change can be measured by rerunning an estimator with the effect altered, the measured value is quoted; where it requires a separate estimate, the estimate is given in one line. A1–A5 concern the flight model, A6–A7 the data, and A8 the definition of a goal. Several of these assumptions are sloppy in the technical sense: the prediction depends on combinations of parameters that the data leave almost free, so a statement about a parameter is not by itself a statement about the prediction [5].

A1 Rigid, non-deforming ball; no contact with post or net. The model needs a constant radius and cross-section, and it stops at the goal plane, before the net. The Laws fix the circumference at 68 cm to 70 cm and the mass at 410 g to 450 g [23], and the problem's $z_0 = 0.11\,\mathrm{m}$ gives $R = 0.110\,\mathrm{m}$. At 31 m/s the aerodynamic pressure $\frac{1}{2}\rho v^2 \approx 0.6\,\mathrm{kPa}$ is of order one per cent of the 0.6 atm to 1.1 atm internal pressure, so the ball stays spherical to well under a millimetre. A 2 mm error in the effective radius changes the cross-section by $2\Delta R/R \approx 3.6$ per cent and the Magnus force, which carries a further factor R, by $3\Delta R/R \approx 5.5$ per cent; applied to the 0.67 m of entry the transverse forces produce, this is a few centimetres. Contact with the post or the net lies outside the model: it would move the crossing by at least the width of the post, and such trajectories are not described by the free-flight input.

A2 Constant drag coefficient during the flight. The speed falls from 32.6 m/s at the kick to roughly 25 m/s at the plane, a quarter of its value, so the Reynolds number stays above the drag crisis at $\mathrm{Re} \approx 9 \times 10^4$–$1.3 \times 10^5$, where $C_D(\mathrm{Re})$ is flat at 0.17–0.22 [11, 12];

that flatness, not the particular value, is what licenses a single C_D [13]. Rather than fixing C_D, we hold it in a Gaussian prior centred at 0.25 with standard deviation 0.15, which covers 0.10–0.40 at one standard deviation and contains both the 0.17–0.22 plateau and the 0.21 measured on a spinning ball [7]; the data then move all fits of this family into $C_D = 0.32$–0.38, above the plateau; that disagreement is examined below. Shifting the prior median across the plateau moves the entry by about a centimetre, and the reason is structural: drag is antiparallel to the velocity, so at fixed launch velocity it cannot change v_y/v_x, and the lateral entry is $y = 3.217\,\mathrm{m}$ for every C_D. Drag does change the flight time, and through it the height: $z = 1.957\,\mathrm{m}$ with gravity alone against $1.879\,\mathrm{m}$ with drag, a shift of 8 cm.

A3 Transverse force linear in spin, fixed axis, exponential decay. The force is $\boldsymbol{F}_M = \frac{1}{2}\rho A C_{L0} R(\boldsymbol{\omega} \times \boldsymbol{v})$ with $C_L = C_{L0} S_p$, $C_{L0} = 1.0$, a constant spin axis, and $\omega(t) = \omega_0 \mathrm{e}^{-t/\tau}$ with $\tau = 6\,\mathrm{s}$. This is the smallest closure that leaves the three spin components free and keeps the force smooth at zero spin. The linear law is supported by measurement: $C_L = 0.21$, 0.24, 0.29 at $S_p = 0.18$, 0.22, 0.34 gives $C_L/S_p = 1.17$, 1.09, 0.85, so the adopted coefficient lies inside the measured range [7]. The main line does not need the spin: it fits the transverse force directly, and a unit lift slope turns its fitted magnitude into an implied spin of 9.6 rev/s, inside the measured 2 rev/s to 10 rev/s. The earlier spinning-ball control, which did carry a spin vector and an exponential decay, found $52(22)\,\mathrm{rad/s}$ (8.3 ± 3.5 rev/s) with $\tau = 6\,\mathrm{s}$; varying τ over 2 s to 12 s there moved the entry by less than 3 cm, and varying its lift coefficient over 0.7–1.3 moved it by 15 cm. So the magnitude of the transverse force is what the data constrain; the mechanism that produces it is not. That is why the six-parameter fit models the force through two coefficients with a zero-centred prior of standard deviation $0.008\,\mathrm{m}^{-1}$ rather than through a spin vector, and why a non-spinning ball's wake asymmetry [8] is a legitimate member of the same family.

A4 Still air. No wind term is included, because two fixes cannot separate a uniform wind from the launch velocity: the wind enters the drag as a relative velocity and is absorbed, to first order, by a change of $\boldsymbol{v}_0$. Refitting with a 3 m/s crosswind and the launch velocity free moves the entry by 0.007 m, so wind of this strength is inside the model envelope rather than a leading term. At fixed launch velocity a 1 m/s crosswind would displace the entry by about 3 cm, but the refit shows that the launch velocity absorbs almost all of it. A neglected wind is also model discrepancy rather than parameter uncertainty, and discrepancy can bias a prediction without widening its stated interval [6], so it is reported as a systematic and not folded into σ_{pos}.

A5 Ball radius 0.110 m, mass 0.430 kg. The model converts aerodynamic coefficients into accelerations through R and m. The Laws bracket the mass at 410 g to 450 g and the circumference at 68 cm to 70 cm, i.e. $R = 0.108\,\mathrm{m}$ to $0.111\,\mathrm{m}$ [23]. The problem's kick height selects $R = 0.110\,\mathrm{m}$, and we use $m = 0.430\,\mathrm{kg}$, the middle of the legal band. Both drag and Magnus accelerations scale as $1/m$, so the ± 20 g band, ± 4.7 per cent, scales the transverse part of the entry by the same fraction; at 0.67 m that is about 3 cm, 0.15 of the 0.20 m predictive standard deviation. The radius enters more weakly because it also enters the area: the ± 2 mm band changes the accelerations by the few per cent estimated in A1 and the entry by a few centimetres.

A6 Gaussian, independent, isotropic position noise with $\sigma_{\text{pos}} = 5$ mm. The likelihood needs a noise model, and two fixes cannot estimate one. The coordinates are printed to 0.01 m, so half of the last printed digit is the only defensible scale; this is not a camera calibration, and the ±1.5 cm of goal-line technology [2] bounds a certified installation rather than these data. Open tracking is coarser: local cubic fits over nine-frame windows leave residuals of 0.056 m and 0.054 m in x and y for IDSSE and 0.065 m and 0.064 m for Metrica, so a goal-line fix is better than a centimetre and a broadcast feed worse, and the printed precision lies between them. The consequence scales as a second difference: the acceleration uncertainty $\sqrt{6}\,\sigma_{\text{pos}}/\Delta t^2$ is 4.9 m/s^2 at 5 mm and 19.6 m/s^2 at 20 mm, where it exceeds the 16 m/s^2 signal itself, and across the twenty-five single-factor variants, four joint-stress cases and an independent replication, P(clean goal) ranges from 0.735 at 20 mm to 0.9997 at 2.5 mm, while the scenario means stay within $y = 2.687$ m to 3.130 m and inside the model envelope. The noise level is the largest single factor after the force law, and the ranking of the estimators survives the whole range.

A7 The fixes are the ball centre, with no common offset. The inversion differences positions, so a common offset is a systematic, not a random, error, and must be stated separately. The problem's $z_0 = 0.11$ m shows that the coordinates refer to the centre. Because the acceleration is a second difference, a common offset b in both fixes appears in the constant-acceleration fit as $2b(t_2 - t_1)/[t_1 t_2(t_1 - t_2)] \approx -400b$ (with b in metres): a 1 cm offset would be read as 4 m/s^2, a quarter of the 16 m/s^2 that the data imply, and would move the entry by 0.21 m, four-tenths of the 0.54 m width of the literature-prior control. Below 2.5 mm the effect is under 5 cm. It is nevertheless the assumption most worth testing with a third fix or a known launch point, and it motivates the systematic line in the error budget.

A8 Crossing of the ball centre; whole-ball scoring. The crossing time and point are defined by the ball centre reaching $x = 11.0$ m, and a goal is scored when the whole ball has passed inside the frame, so the clean-goal region is $|y| \le 3.66 - 0.11 = 3.55$ m and $0.11 \le z \le 2.44 - 0.11 = 2.33$ m [23]. All probabilities quoted here use this whole-ball convention. The distinction is not cosmetic: it shrinks the effective target from 7.32 m × 2.44 m to 7.10 m × 2.22 m, a loss of 12 % of its area. With the nominal predictive standard deviations $\sigma_y \approx \sigma_z \approx 0.20$ m, the predicted centre lies 3.7σ inside the post and 1.9σ below the crossbar; removing one radius from each side lowers P(clean goal) from 0.993 under a centre-only convention to 0.971, a difference of two percentage points that comes almost entirely from the crossbar. A centre-only convention is not directly comparable with the numbers reported here.

4 Physical Model

4.1 Coordinates, the target plane, and the observations

We place the origin at the centre of the penalty mark, with x horizontal towards the centre of the goal, y across the goal, and z vertically upwards (Fig. 1). The ball is struck at $t = 0$ with its centre at $\boldsymbol{r}_0 = (0, 0, 0.11)$ m, one radius above the ground. The goal is the plane $x = x_g = 11.0$ m, whose opening is $-b \le y \le b$ with $b = 3.66$ m and $0 \le z \le h$ with $h = 2.44$ m, a rectangle of 7.32 m by 2.44 m [23]. A goal counts only if the whole ball is past the line and inside the frame. For a sphere of radius $R = 0.110$ m this requires the centre at the

crossing to satisfy

$$|y| \le b - R = 3.55 \text{ m}, \qquad R \le z \le h - R = 2.33 \text{ m}, \tag{2}$$

because the sphere meets the plane in a disc of radius R. All probabilities in this paper use this whole-ball convention, which treats the goal as a plane of zero thickness and ignores contact with, or deformation of, the frame.

The method receives two measured positions of the ball centre, $\boldsymbol{r}_1 = (1.50, 0.45, 0.44)$ m at $t_1 = 0.05$ s and $\boldsymbol{r}_2 = (2.96, 0.86, 0.74)$ m at $t_2 = 0.10$ s (Fig. 1), and returns the point where the trajectory crosses $x = x_g$. At the fitted launch speed the crossing occurs at $t = 0.3988$ s, so the fixes sample the first quarter of the flight and everything beyond 0.10 s is extrapolation. The task is therefore not interpolation between fixes but out-of-sample continuation of a trajectory whose aerodynamic force is itself unmeasured.

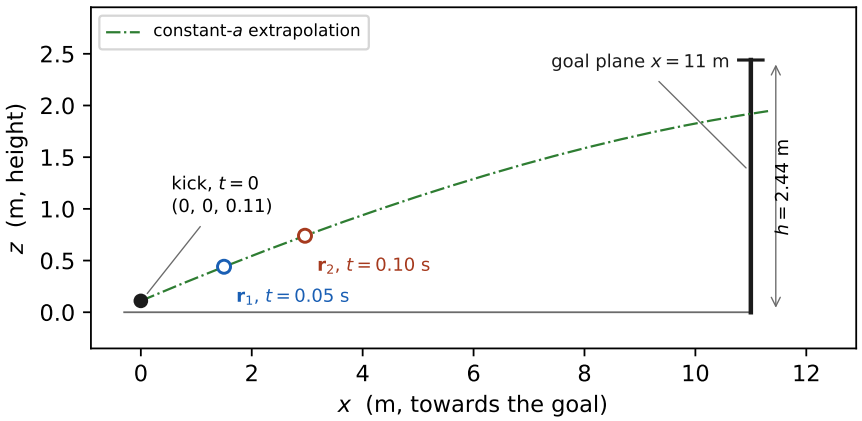


Figure 1: Geometry of the problem in the side view. The goal is the plane $x = 11$ m with an opening 7.32 m wide (not visible in this view) and 2.44 m high. The filled circle is the kick point, the two open circles are the given observations $\boldsymbol{r}_1$ at 0.05 s and $\boldsymbol{r}_2$ at 0.10 s, and the dash-dot line is the constant-acceleration extrapolation retained as a control in §6.

4.2 Equations of motion and a transverse-force model

We model the ball as a rigid sphere of mass $m = 0.430$ kg and radius $R = 0.110$ m, with cross-section $A = \pi R^2 = 0.038013$ m^2, moving through air of density $\rho = 1.20$ kg m^{-3} that may itself move with velocity $\boldsymbol{w}$. Let $\boldsymbol{u} = \boldsymbol{v} - \boldsymbol{w}$ be the air velocity relative to the ball. The equations of motion are

$$\frac{\mathrm{d}\boldsymbol{v}}{\mathrm{d}t} = -\mathrm{FACTOR}\, C_D\, |\boldsymbol{u}|\, \boldsymbol{u} + |\boldsymbol{u}|\, (\boldsymbol{\beta} \times \boldsymbol{u}) - g\, \hat{\boldsymbol{z}}, \qquad \frac{\mathrm{d}\boldsymbol{r}}{\mathrm{d}t} = \boldsymbol{v}, \tag{3}$$

where $g = 9.81$ m s^{-2}, $\boldsymbol{u} = \boldsymbol{v} - \boldsymbol{v}_{\mathrm{wind}}$, $\mathrm{FACTOR} = \rho A/(2m) = 0.053042$ m^{-1}, and $\boldsymbol{\beta}$ is a constant vector with units m^{-1}.

The first term is quadratic drag along $-\boldsymbol{u}$ [13, 24]. Being parallel to the relative velocity, it changes the speed but not the direction of motion in still air. The second term is the transverse force: its magnitude is at most $|\boldsymbol{\beta}||\boldsymbol{u}|^2$ and its direction $\boldsymbol{\beta} \times \boldsymbol{u}$ is perpendicular to $\boldsymbol{u}$, so it does no work in still air. This is the form of the force on a spinning ball, for which $\boldsymbol{\beta}$ points along the spin axis and its magnitude is proportional to the spin at a given speed, and

it also represents the side force on a non-spinning ball whose wake is asymmetric [8], since for any direction perpendicular to $\boldsymbol{u}$ a suitable $\boldsymbol{\beta}$ produces it. The fit therefore determines the force itself rather than the spin, and presupposes neither mechanism.

To expose the two independent directions of the force, we resolve $\boldsymbol{\beta}$ in a basis attached to the launch direction. Let $\boldsymbol{u}_0 = \boldsymbol{v}_0 - \boldsymbol{w}$, $\boldsymbol{e}_0 = \boldsymbol{u}_0/|\boldsymbol{u}_0|$, $\boldsymbol{e}_1 = (\hat{\boldsymbol{z}} \times \boldsymbol{e}_0)/|\hat{\boldsymbol{z}} \times \boldsymbol{e}_0|$, which is horizontal and perpendicular to the flight plane, and $\boldsymbol{e}_2 = \boldsymbol{e}_0 \times \boldsymbol{e}_1$, which lies in that plane. We write

$$\boldsymbol{\beta} = \beta_1 \boldsymbol{e}_1 + \beta_2 \boldsymbol{e}_2, \tag{4}$$

so that, by the cross product, β_2 drives the sideways force and β_1 the lift. The component of $\boldsymbol{\beta}$ along $\boldsymbol{e}_0$ is omitted: at $t = 0$ it produces no force, because $\boldsymbol{\beta} \times \boldsymbol{u}_0 = 0$, and later it produces only a small force, because the relative velocity turns by a few degrees during the flight. Section 5 shows that this component is also the one the data cannot determine. The flight is then described by six parameters,

$$\boldsymbol{\theta} = (\boldsymbol{v}_0,\, C_D,\, \beta_1,\, \beta_2), \tag{5}$$

the launch velocity, the drag coefficient, and the two transverse coefficients. The nominal analysis takes still air, $\boldsymbol{w} = 0$; uniform wind is reinstated as a scenario in §5, where it enters only through $\boldsymbol{u}$.

4.3 Aerodynamic scales and the fitted coefficients

The dimensionless groups are the Reynolds number $Re = U(2R)/\nu$ and the spin parameter $S_p = \omega R/U$, with $U = |\boldsymbol{u}|$ the relative speed and $\nu = 1.5 \times 10^{-5}$ m^2 s^{-1} the kinematic viscosity. At the penalty speed $U \approx 31$ m s^{-1}, $Re \approx 4.4 \times 10^5$. The drag crisis of a football occurs at $Re \approx$ (9–13) $\times\, 10^4$, that is $U \approx$ 6–9 m s^{-1}; above it C_D is nearly flat at 0.17–0.22 [11, 12]. The flight therefore lies well inside the flat region, so a constant C_D is a good approximation; the uncertainty is the ball-to-ball scatter of its level, not the shape of $C_D(Re)$. That level is also difficult to recover from a short trajectory: whole-trajectory estimates of C_D and C_L can disagree with estimates from the local curvature and torsion of the track [15], and trajectory-based drag estimates are sensitive to the assumed force model [14]. We therefore fit C_D with a broad prior rather than fixing it from wind-tunnel data.

The transverse force is constrained by the same two fixes. Measurements give lift coefficients $C_L = 0.21$–0.29 at $S_p = 0.18$–0.34 [7], consistent with the linear law $C_L = C_{L0} S_p$ and C_{L0} of order unity. In the present parameterisation the equivalent lift coefficient is $C_L = |\boldsymbol{\beta}|/\mathrm{FACTOR}$, which writes the transverse acceleration as $C_L \rho A |\boldsymbol{u}|^2/(2m)$. The fitted value, $C_L = 0.159$ (§5), lies just below the measured range. Under the measured linear law it is the force on a ball spinning at several revolutions per second and corresponds to a transverse acceleration of the same order as gravity, the scale on which a spinning or seam-oriented ball curves; it moves the crossing by a distance large compared with the precision of the fixes. Table 2 collects the constants and Fig. 2 places the flight relative to the drag crisis and the lift measurements.

4.4 Drag alone cannot bend the shot

The model has a structural property that organises the analysis. With drag and gravity only, and still air, the horizontal equation is $\dot{\boldsymbol{v}}_\perp = -k|\boldsymbol{v}|\boldsymbol{v}_\perp$ with $k = \mathrm{FACTOR}\, C_D > 0$. Both

Table 2: Constants of the flight model. The last column gives the source of each value or the rule that fixes it. The position precision is a property of the printed coordinates, not a calibration of a tracking system.

Quantity	Symbol	Value used	Source or rule
Air density (20 °C, sea level)	ρ	1.20 kg/m^3	$\pm 4\%$
Kinematic viscosity	ν	1.5×10^{-5} m^2/s	air at 20 °C
Ball radius	R	0.110 m	circumference 68–70 cm [23]
Ball mass	m	0.430 kg	410–450 g [23]
Cross-section	A	3.8013×10^{-2} m^2	πR^2
Gravity	g	9.81 m/s^2	standard value
Drag factor	FACTOR	5.3042×10^{-2} m^{-1}	$\rho A/(2m)$
Position precision	σ_{pos}	5 mm	half the last printed digit

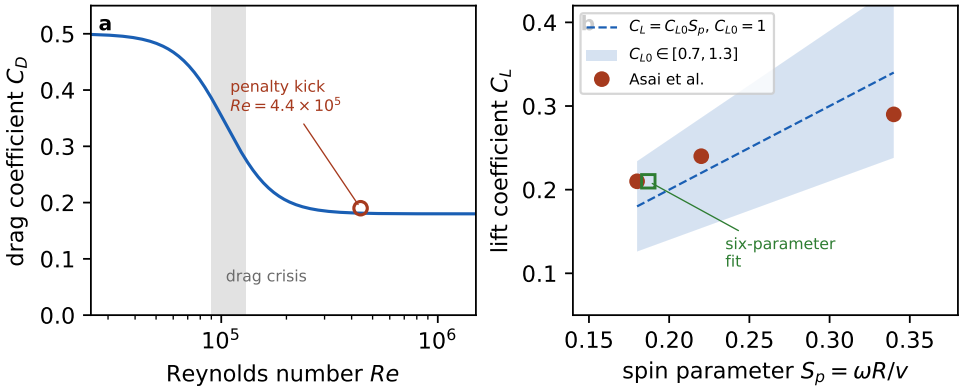


Figure 2: Aerodynamic inputs. (a) The drag crisis of a football, shown as a guide to the eye through the values of [11, 12]: C_D falls through $Re \approx (9\text{–}13) \times 10^4$ and is flat above it. The open circle is the operating point of a penalty kick at 31 m/s. (b) Lift coefficient against spin parameter: the model's $C_L = C_{L0} S_p$ with $C_{L0} = 1$ and its 0.7–1.3 band (dashed and shaded), the measurements of [7] (filled circles), and the spin parameter of the earlier spinning-ball fit retained as a control (§6).

components of $\boldsymbol{v}_\perp$ are multiplied by the same positive scalar, so their ratio is conserved:

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{v_y}{v_x}\right) = 0 \quad \Longrightarrow \quad \frac{y(t)}{x(t)} = \frac{v_{0y}}{v_{0x}} \quad \Longrightarrow \quad y_{\text{entry}} = x_g \, \frac{v_{0y}}{v_{0x}}, \tag{6}$$

independently of the magnitude of the drag and, indeed, of any drag law that depends only on speed. Drag changes the flight time and the height at the plane, but it cannot move the crossing sideways relative to the initial aim; only a force with a component perpendicular to $\boldsymbol{v}$ can. This is why the transverse coefficients, not C_D, control the lateral prediction.

The data show the same structure. A drag-and-gravity model fitted to the six numbers returns $y = 3.2172$ m at the plane, whereas the exact constant-acceleration fit of the same numbers returns $y = 2.4948$ m. Both reproduce the fixes to within their precision, which is precisely the difficulty: two fixes alone can barely separate the two descriptions, and no choice of C_D can close the gap, because the drag-only crossing is tied to $x_g v_{0y}/v_{0x}$ by Eq. (6). A uniform wind breaks the identity, since drag is then along $\boldsymbol{u}$ rather than $\boldsymbol{v}$, but over the 0.1 s

baseline it is nearly degenerate with the launch velocity and the refit absorbs most of it (§5). The lateral answer is therefore an inference, not a measurement.

4.5 Numerical integration and the target-plane crossing

The state $\boldsymbol{s} = (\boldsymbol{r}, \boldsymbol{v})$ is advanced with the classical fourth-order Runge–Kutta scheme

$$\boldsymbol{s}_{n+1} = \boldsymbol{s}_n + \frac{\Delta t}{6}\left(\boldsymbol{f}_1 + 2\boldsymbol{f}_2 + 2\boldsymbol{f}_3 + \boldsymbol{f}_4\right) + \mathcal{O}\left(\Delta t^5\right), \tag{7}$$

at the fixed step $\Delta t = 0.005$ s. The crossing of $x = x_g$ is located by cubic Hermite interpolation of $x(t) - x_g$ across the bracketing step, using the position and velocity supplied at both ends; since the interpolant matches both values and both derivatives, its error is $\mathcal{O}(\Delta t^4)$, the same order as the integration. Halving Δt changes the entry ordinates by less than 1 mm, far below the 5 mm precision of the fixes and the spread of the prediction. The same integrator and crossing rule are used in the fit and in the prediction, so no interpolation of the model enters between the two stages.

5 Method: From Two Fixes to a Predictive Distribution

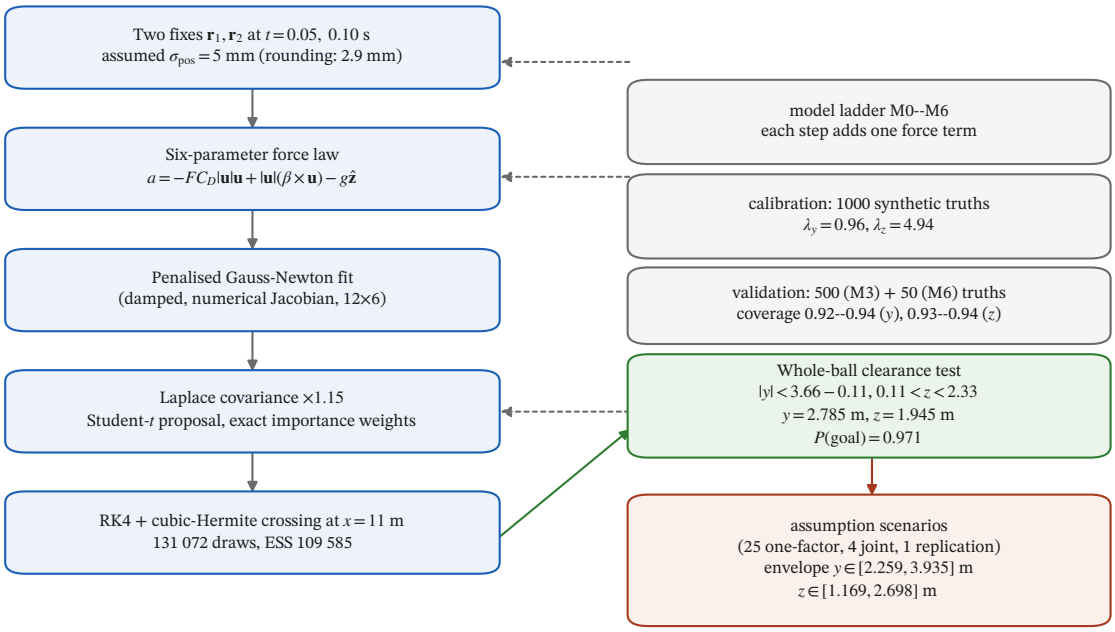


Figure 3: The prediction pipeline of Sections 4–6, from the two fixes to the reported interval and probability. Each stage is checked against an independent calculation: the integrator against DOP853, the Laplace covariance against a second sampler, and the whole chain against 750 synthetic truths: 500 on M3, 50 on M6 and 25 under a law containing an extra vertical force.

5.1 The inverse problem and the noise model

The data are the six coordinates of $\boldsymbol{r}_1$ and $\boldsymbol{r}_2$. The forward map $\boldsymbol{F}(\boldsymbol{\theta}) = (\boldsymbol{r}(t_1;\boldsymbol{\theta}), \boldsymbol{r}(t_2;\boldsymbol{\theta}))$ carries the six parameters of Eq. (5) to the two predicted positions, and the inverse problem is to characterise the set of $\boldsymbol{\theta}$ consistent with the data. A point inversion is inappropriate for two

reasons. The map is nonlinear, so even with as many data as parameters the solution need not be unique; and the model is incomplete, because a non-spinning ball can curve through wake asymmetry [8] and the effective force may vary from ball to ball. The fitted $\boldsymbol{\beta}$ is therefore an effective coefficient, not a spin measurement. The same inverse problem has been solved for baseball from a large set of tracked pitches, where the data could separate drag, lift, and the spin vector [16]; two fixes cannot support that separation, and the prior must carry part of the inference. Estimates of aerodynamic coefficients from a trajectory are also known to depend on the estimator [15], and different conventions for reconstructing spin give different answers [17]; a probabilistic treatment makes that dependence explicit instead of hiding it.

We adopt the measurement model

$$\boldsymbol{r}_i^{\mathrm{obs}} = \boldsymbol{r}(t_i;\boldsymbol{\theta}) + \boldsymbol{\varepsilon}_i, \qquad \boldsymbol{\varepsilon}_i \sim \mathcal{N}(\mathbf{0}, \sigma_{\mathrm{pos}}^2 \boldsymbol{I}_3), \tag{8}$$

in which each coordinate is observed with independent Gaussian error of standard deviation $\sigma_{\mathrm{pos}} = 5$ mm. This value is half of the last printed digit of the given coordinates, which are quoted to 1 cm; it is a statement about the printed data, not a calibration of any tracking system, and the problem supplies no measurement error. Goal-line technology is certified to about 1.5 cm [2], optical systems in two dimensions only [4], and a three-dimensional fix normally requires several calibrated views [3]; single-camera reconstruction remains a research problem [19]. The assumed 5 mm is therefore optimistic, and §6.6 repeats the inference with σ_{pos} between 2.5 and 20 mm.

5.2 Information content of six coordinates

Before fitting, we ask which combinations of parameters the data constrain. Consider the seven-parameter model in which $\boldsymbol{\beta}$ is a free vector. Its data Jacobian $\boldsymbol{J} = \partial \boldsymbol{F}/\partial \boldsymbol{\theta}_7$, with every row divided by σ_{pos}, is 6×7 and has rank 6: the observations are blind to one direction in parameter space. The six nonzero singular values are 327.227, 318.306, 217.813, 1.766, 1.765, and 1.750. The three large values describe the launch geometry; the three values near 1.75 describe the aerodynamic coefficients and are almost degenerate. The blind direction is dominated by a joint change of the components of $\boldsymbol{\beta}$ and contains the component along the instantaneous air velocity: when $\boldsymbol{\beta} \parallel \boldsymbol{u}$ the force $|\boldsymbol{u}|(\boldsymbol{\beta} \times \boldsymbol{u})$ vanishes, so that component produces no acceleration at first order and the data cannot see it. In the seven-parameter posterior the variance of this component shrinks by only 8.69% relative to the prior, against 69.99% and 72.65% for the two transverse components; a local information count gives about 5.27 data-determined parameters. This is the concrete form of a sloppy model [5]: a spectrum spanning two orders of magnitude, with the flat directions fixed not by data but by the prior [6].

The six-parameter model of Eq. (5) removes the blind direction by construction: it sets the along-flight component to zero and fits the two components that produce a force. This is the largest model the data can support, and its posterior is conditional on that choice. The cost of the choice can be measured. In the seven-parameter parameterisation, fixing the lab-frame component closest to the flight direction and scanning it over ± 0.075 m^{-1}, with the remaining components refitted at each point, moves the predicted crossing from $y = 2.555$ m to 2.782 m and from $z = 2.116$ m to 1.849 m. The changes, 0.228 m in y and 0.267 m in z, are comparable to the predictive standard deviation and far larger than the precision of the fixes. The seven-parameter model is rank-deficient: six observations cannot identify seven

Table 3: Priors and hard bounds on the six parameters. The priors are independent normals and the bounds truncate them.

Parameter	Prior	Hard bounds
v_{0x}	$\mathcal{N}(30,\, 15^2)$ m/s	5–60 m/s
v_{0y}	$\mathcal{N}(0,\, 15^2)$ m/s	−30–30 m/s
v_{0z}	$\mathcal{N}(7,\, 10^2)$ m/s	0–30 m/s
C_D	$\mathcal{N}(0.25,\, 0.15^2)$	0–1
β_1	$\mathcal{N}(0,\, 0.008^2)$ m^{-1}	± 0.08 m^{-1}
β_2	$\mathcal{N}(0,\, 0.008^2)$ m^{-1}	± 0.08 m^{-1}

parameters, and its posterior is a conditional statement, not a measurement. We therefore report the six-parameter posterior and pair it with a model envelope rather than a single interval.

5.3 Likelihood and priors

With the noise model of Eq. (8), the likelihood is

$$p(\boldsymbol{r}_1, \boldsymbol{r}_2 \mid \boldsymbol{\theta}) \propto \exp\left[-\frac{1}{2\sigma_{\text{pos}}^2} \sum_{i=1}^{2} \left\| \boldsymbol{r}(t_i; \boldsymbol{\theta}) - \boldsymbol{r}_i \right\|^2 \right], \tag{9}$$

truncated by the hard bounds below. The priors are independent normals: $\boldsymbol{v}_0 \sim \mathcal{N}((30, 0, 7), \mathrm{diag}(15^2, 15^2, 10^2))$ m s^{-1}, $C_D \sim \mathcal{N}(0.25, 0.15^2)$, and $\beta_i \sim \mathcal{N}(0, 0.008^2)$ m^{-1} for $i = 1, 2$, with the hard bounds $5 < v_{0x} < 60$ m s^{-1}, $|v_{0y}| < 30$ m s^{-1}, $0 < v_{0z} < 30$ m s^{-1}, $0 < C_D < 1$, and $|\beta_i| < 0.08$ m^{-1}; Table 3 lists them. The launch prior is broad compared with the precision of the fixes, so the data, not the prior, fix the launch direction. The drag prior is wider than the wind-tunnel range 0.17–0.22 [11, 12], because the effective drag of a spinning ball need not match the tabulated value. The transverse prior is centred at zero: it expresses no preference for left or right curl and no assumption that the force is Magnus. Its scale, 0.008 m^{-1}, corresponds through $C_L = |\boldsymbol{\beta}|/\mathrm{FACTOR}$ to lift coefficients of the measured order, so it admits the observed forces while remaining weak along the two determined directions. A prior on the spin would instead introduce a particular force law that the data cannot test [8]; parametrising the force directly is the more honest choice.

5.4 Posterior computation

We combine likelihood and priors with Bayes' rule and compute the posterior in two stages: a Laplace approximation locates and scales the target, and exact importance sampling corrects it. The joint residual is

$$\boldsymbol{R}(\boldsymbol{\theta}) = \left(\frac{\boldsymbol{r}(t_1; \boldsymbol{\theta}) - \boldsymbol{r}_1}{\sigma_{\text{pos}}},\; \frac{\boldsymbol{r}(t_2; \boldsymbol{\theta}) - \boldsymbol{r}_2}{\sigma_{\text{pos}}},\; \frac{\boldsymbol{\theta} - \boldsymbol{\mu}}{\boldsymbol{s}} \right), \tag{10}$$

a vector of length twelve whose first six entries are the data residuals and whose last six are the prior pulls, with $\boldsymbol{\mu}$ and $\boldsymbol{s}$ the prior means and standard deviations. The MAP estimate minimises $\tfrac{1}{2}\|\boldsymbol{R}\|^2$ under the hard bounds by a damped Gauss–Newton iteration with a finite-difference Jacobian and adaptive damping, so no derivative of the integrator is required. The

solution has $C_D = 0.3466$, $|\boldsymbol{\beta}| = 0.008426$ m^{-1} (equivalently $C_L = 0.159$), and crosses the plane at $y = 2.7830$ m, $z = 1.9451$ m, $t = 0.3988$ s. At the mode, the Gauss–Newton approximation to the inverse Hessian of $\frac{1}{2}\|\boldsymbol{R}\|^2$ gives the Laplace covariance

$$\boldsymbol{C} = 1.15\left(\boldsymbol{J}^{\mathsf{T}}\boldsymbol{J}\right)^{-1}, \tag{11}$$

where $\boldsymbol{J}$ is the 12×6 Jacobian of $\boldsymbol{R}$ at the MAP and the factor 1.15 inflates the local quadratic approximation.

The Laplace covariance is not a profile-likelihood interval: it is symmetric by construction and uses the curvature at the mode, whereas a profile interval follows the likelihood along one parameter after maximising over the others, can be asymmetric, and along the ridge of §5.2 can differ substantially. We therefore use the Laplace covariance only as a proposal. Draws are taken from a multivariate Student-t distribution with eight degrees of freedom, centred at the MAP with scale matrix $\boldsymbol{C}$; the heavy tails ensure that the proposal does not underestimate the tails of the target. Each draw $\boldsymbol{\theta}_j$ carries the exact importance weight

$$w_j \propto \frac{p(\boldsymbol{r}_1, \boldsymbol{r}_2 \mid \boldsymbol{\theta}_j)\, p(\boldsymbol{\theta}_j)}{q(\boldsymbol{\theta}_j)}, \tag{12}$$

where q is the proposal density. Because the weights are exact, the weighted draws represent the posterior itself, not its Laplace approximation, and the reported intervals are weighted quantiles. We draw 131072 samples and retain an effective sample size of 109585, a ratio that indicates the proposal is close to the target. A Markov chain would also sample the posterior, but it would require convergence diagnosis along a ridge and a separate run for every scenario; independent weighted draws make the twenty-six cases of §6.6 directly comparable.

The computation was cross-checked: an independent parameterisation of the transverse force (three lab-frame components instead of two transverse ones), an independent sampler of 40000 draws, and an independent reference run of 131072 draws all reproduce the prediction; an adaptive DOP853 integration agrees with the RK4 crossing to better than 2×10^{-6} m; and the posterior mean, $(2.7847, 1.9452)$ m, differs from the MAP by 1.4 mm in y and 1.0 mm in z.

5.5 Prediction and goal probability

Every weighted draw is propagated through the full nonlinear equations to $x = x_g$ with the RK4 scheme and crossing rule of §4; the map from parameters to entry point is never linearised. This matters because impact-point distributions can be markedly non-Gaussian when the entry state and the aerodynamic parameters are uncertain [22], and because propagating uncertain aerodynamic coefficients to impact dispersion is the standard use of such fits [21]. The predictive distribution has mean $y = 2.7847$ m with 90% interval $[2.45, 3.12]$ m, mean $z = 1.9452$ m with 90% interval $[1.62, 2.27]$ m, and mean crossing time 0.399 s. A negligible fraction of the draws does not reach the plane and is excluded, so the distribution is conditional on reaching it.

A goal is decided for the whole ball by Eq. (2): the crossing is successful if the disc of radius R lies inside the aperture. The goal probability is the weighted fraction of draws satisfying this criterion,

$$P(\text{goal}) = 0.971 \pm 0.001, \tag{13}$$

where the quoted uncertainty is the Monte Carlo standard error of the weighted estimate; it measures sampling error, not model error. The assumption-driven uncertainty is quantified separately in §6.6.

5.6 Assumption scenarios

To measure how much the answer depends on assumptions that two fixes cannot test, the entire inference — refit, importance weights, and propagation — is repeated under twenty-five single-factor variants of the nominal assumptions, under four joint-stress combinations, and in one independent replication. The variants are: $\sigma_{\text{pos}} = 2.5$, 10, and 20 mm; a correlation of -0.5 or 0.7 between the errors of the same coordinate at the two epochs; an epoch jitter of 0.5 or 2 ms, propagated as a position uncertainty along the instantaneous velocity; a transverse prior width multiplied by 0.5 or 2; the prior mean of β_1 or β_2 shifted by one standard deviation in either direction; a drag prior mean of 0.1 or 0.4; a uniform wind of 3 m s^{-1} along each axis in both directions; a transverse force with a component along the launch direction, $\gamma = \pm 0.016$ m^{-1}; and a drag coefficient with a strong linear dependence on speed, of either sign.

Across these cases the goal probability ranges from 0.735 at $\sigma_{\text{pos}} = 20$ mm to 0.9997 at 2.5 mm, the scenario means from 2.687 to 3.130 m in y and 1.846 to 2.042 m in z, and the union of the 95% marginal intervals is $[2.259, 3.935]$ m in y and $[1.169, 2.698]$ m in z. Four additional joint-stress cases, which combine several adverse variants, give goal probabilities from 0.561 to 0.987 and means from 2.733 to 3.339 m in y. A uniform wind is nearly a Galilean shift of the launch velocity over the short baseline, so the refit absorbs most of it and a 3 m s^{-1} crosswind moves the entry by only 0.007 m. The two extremes of the goal probability are produced by the fix precision, not by the aerodynamic model.

This is a discrete scenario envelope, not a confidence or credible interval. The cases are selected rather than sampled from a distribution over assumptions, they are not equally probable, and the envelope is not a guaranteed bound. It answers a narrower question: how far can the prediction move when one assumption at a time is changed by an amount that a careful analyst might defend? The full set is reported in §6.

5.7 Control estimators

The same six numbers were analysed earlier with two simpler estimators: a constant-acceleration inversion that fits the two fixes exactly, and a drag-plus-Magnus Bayesian fit whose spin prior came from the literature. Both are retained as controls and quantified in §6; neither is used for the headline prediction.

6 Results

This section reports what the two fixes at $t = 0.05$ and $t = 0.10$ s imply for the crossing of the plane $x = 11$ m. We first extract the acceleration and the force balance that the fixes determine, then compare seven models of increasing physical content, examine the statistical intervals and their calibration, audit the identifiability of the fitted parameters, and give the posterior answer. Three further subsections state the assumption-scenario envelope, the validation of the whole pipeline on synthetic truths, and a design study of what additional tracking would buy. The earlier constant-acceleration and literature-prior estimators are reported once, at the end, as control models. Unless stated otherwise, all uncertainties refer to the adopted position noise $\sigma_{\text{pos}} = 5\,\text{mm}$, which is half of the last printed digit of the given

coordinates and is an assumption about the data, not a camera calibration.

6.1 Acceleration and force implied by the fixes

The two measured positions contain six independent coordinates after the kick. A constant acceleration $\boldsymbol{a}$ and an initial velocity $\boldsymbol{v}_0$ reproduce all six exactly, and continuing the resulting parabola to $x = 11$ m gives

$$\boldsymbol{a} = (-16.000,\ -16.000,\ -12.000)\ \mathrm{m\,s^{-2}}, \qquad \boldsymbol{v}_0 = (30.4,\ 9.4,\ 6.9)\ \mathrm{m\,s^{-1}}, \tag{14}$$

with $|\boldsymbol{a}| = 25.6\ \mathrm{m\,s^{-2}}$ and a flight time of 0.405008 s to the plane. Because there are six data and six unknowns, this interpolation has no residual and no freedom: the six numbers are a kinematic measurement, not a model assumption, and their only uncertainty is the one inherited from σ_{pos}. The magnitude is already informative. It is 2.6 times g, so the ball is not in free fall, and its vertical component, $-12.0\ \mathrm{m\,s^{-2}}$, differs from $-g$ by $2.19\ \mathrm{m\,s^{-2}}$. The horizontal components are equal, $a_y/a_x = 1.00$, whereas pure drag along a velocity with $v_{0y}/v_{0x} \approx 0.31$ would give a horizontal acceleration ratio of only about 0.31, a factor of three smaller. The measured acceleration therefore contains a lateral force that gravity and drag cannot supply.

Subtracting gravity isolates the aerodynamic part,

$$\boldsymbol{a}_{\mathrm{aero}} = (-16.000,\ -16.000,\ -2.190)\ \mathrm{m\,s^{-2}},$$

whose magnitude is $22.7\ \mathrm{m\,s^{-2}}$, larger than g. At the first fix the speed is $31.5\ \mathrm{m\,s^{-1}}$. Resolving $\boldsymbol{a}_{\mathrm{aero}}$ parallel and perpendicular to the velocity gives 19.87 and 11.05 $\mathrm{m\,s^{-2}}$ respectively. The parallel part determines an effective drag coefficient $C_D = 0.378$, the perpendicular part an effective lift coefficient $C_L = 0.211$, and the perpendicular direction fixes the spin axis $\hat{\boldsymbol{\omega}} = (0.2357,\ -0.1035,\ -0.9663)$. At a unit lift slope $C_{L0} = 1$ this axis and coefficient correspond to $9.58\ \mathrm{rev\,s^{-1}}$. Table 4 collects the decomposition, and Fig. 4 shows the two projections of the trajectory and the extrapolation that the measured acceleration produces.

The two coefficients are effective whole-flight values, and their comparison with independent measurements is instructive rather than reassuring. The drag coefficient 0.378 is roughly twice the wind-tunnel value of 0.19–0.22 for a football at these speeds [11, 12], while the lift coefficient 0.211 is large. If the whole perpendicular force were Magnus with a lift slope near unity, the required spin would be $9.58\ \mathrm{rev\,s^{-1}}$, which is at the top of the 2–10 $\mathrm{rev\,s^{-1}}$ range measured for penalties [7, 9]; a lower lift slope would demand proportionally more spin. A lateral force without spin is, however, documented for knuckleballs, where wake asymmetry rather than the Magnus effect produces the deflection [8]. Two fixes cannot separate these mechanisms, so the perpendicular component is best read as an effective transverse force, a point to which we return when the six-parameter model is defined.

6.2 The model ladder

A single point prediction hides the fact that different admissible forces give different answers. We therefore fit a ladder of seven models, each adding one piece of physics, and record where each model puts the crossing. The models, their parameter counts, residuals and entry points are listed in Table 5 and drawn in Fig. 6. The residual is the root-mean-square of the six coordinate residuals; at $\sigma_{\mathrm{pos}} = 5$ mm a residual of 5 mm is one noise standard deviation per

Table 4: Force balance at the first fix, from the exact constant-acceleration fit. The aerodynamic vector is the measured acceleration minus gravity. Its decomposition parallel and perpendicular to the velocity gives the effective coefficients in the last column (three significant figures in the text: $C_D = 0.378$, $C_L = 0.211$). The perpendicular direction is the spin axis $\hat{\boldsymbol{\omega}} = (0.2357, -0.1035, -0.9663)$, i.e. 9.58 rev s^{-1} at $C_{L0} = 1$.

Quantity	Vector (m s^{-2})	Magnitude (m s^{-2})	Derived
Measured $\boldsymbol{a}$	$(-16.000, -16.000, -12.000)$	25.6	
Gravity $\boldsymbol{a}_g$	$(0, 0, -9.81)$	9.81	
Aerodynamic $\boldsymbol{a} - \boldsymbol{a}_g$	$(-16.000, -16.000, -2.190)$	22.7	
Parallel to $\boldsymbol{v}$	—	19.87	$C_D = 0.3784$
Perpendicular to $\boldsymbol{v}$	—	11.05	$C_L = 0.2105$

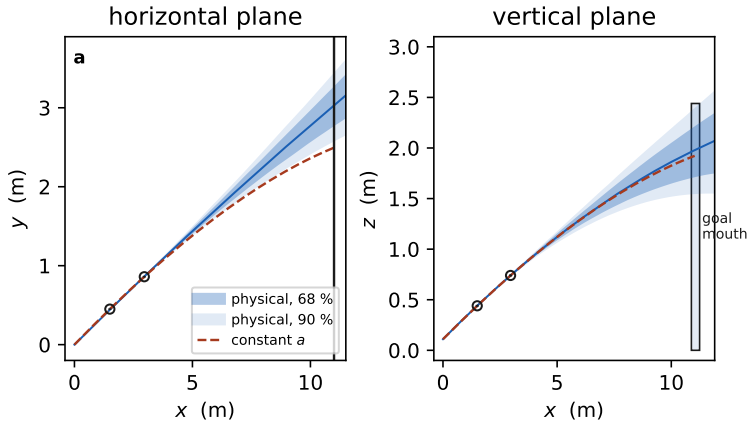


Figure 4: Ball trajectory through the three sampled positions projected onto the horizontal (a) and vertical (b) planes, with two continuations to the goal plane: the constant-acceleration extrapolation (dashed) and a drag-and-spin continuation with its 68% and 90% bands (shaded). The two continuations agree through the fixes and separate in the extrapolation; the shaded bands belong to the earlier control analysis and are shown only to indicate the extrapolation uncertainty.

coordinate.

M0 is the exact interpolation of Eq. (14): six parameters, zero residual, entry $(2.4948, 1.9204)$ m. It is not a force model, but it is the correct top rung because it shows what the data alone say before any dynamics is imposed. M1 replaces the measured acceleration by gravity alone, leaving only the three launch-velocity components. Its residual rises to 10.376 mm, about two standard deviations per coordinate, and its entry moves to $(3.2170, 1.9570)$ m: without drag or lift the ball keeps the horizontal direction it had at the first fix, and the prediction misses by 0.72 m in y. M2 adds quadratic drag with one coefficient. The residual halves to 5.029 mm, but the entry moves by only 0.2 mm in y and -78 mm in z. This is exactly what drag along the velocity should do: it changes the deceleration and hence the vertical drop, but it cannot change the ratio v_y/v_x that sets the lateral position. Drag buys vertical realism and nothing lateral.

M3 adds a constant lateral acceleration $a_y = -11.422$ m s^{-2}. The residual collapses by a factor 9.3 to 0.542 mm, and the entry moves 0.67 m left to $(2.5486, 1.8931)$ m. The improvement is not a marginal statistical preference. The M2 profile interval in y, $[3.197, 3.237]$ m, is

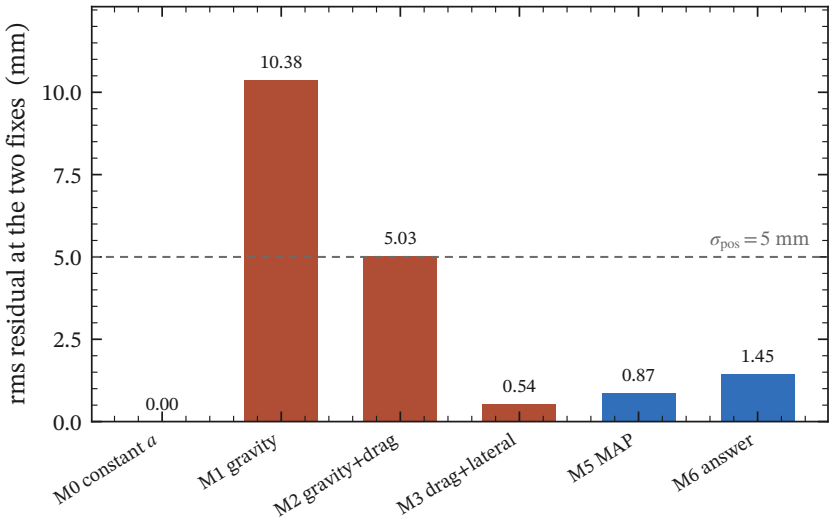


Figure 5: Goodness of fit at the two fixes for the nested models of Table 5, in units of the assumed position noise $\sigma_{\text{pos}} = 5\,\text{mm}$ (dashed line). Gravity alone misses by more than two noise widths; drag alone still misses by one; adding a lateral force term collapses the residual by an order of magnitude, and the six-parameter answer fits within the noise. The ladder therefore measures how many force terms the data support, not how well one model was tuned.

disjoint from the M3 interval $[2.144, 2.954]$ m, so the two fixes demand a lateral force at high significance. With five parameters (three for $\boldsymbol{v}_0$, one for drag, one for the lateral acceleration) M3 is the most economical description that fits, and it is the best two-parameter compromise among the physical models: one coefficient for the force along the flight and one for the force across it. M4 keeps the same two forces but derives them instead of fitting them: it reads drag and Magnus off the M0 acceleration at the first fix and therefore has no free parameters at all. Its entry, $(2.6481, 1.9568)$ m at $t = 0.4007$ s, lies between M3 and the prior-weighted models. M5 is the seven-parameter joint Bayesian fit with literature priors, quoted only as a prior-sensitivity aside; its MAP entry is $(2.7073, 1.9541)$ m with a residual of 0.869 mm. M6 is the six-parameter transverse-force model used for the answer: it retains drag and two coefficients for the force perpendicular to the flight, drops the along-flight spin component that the data cannot see, and gives a posterior mean entry of $(2.7847, 1.9452)$ m at $t = 0.3988$ s with a residual of 1.448 mm. Its larger residual relative to M5 is the price of the constrained, physical parameterisation and of the priors; it is still well inside the noise.

The ladder's message is the spread, not any single rung. The entry ordinate ranges from 2.4948 to 3.2170 m, a spread of 0.72 m, while the entry height ranges only from 1.879 to 1.957 m, a spread of 0.08 m. At $\sigma_{\text{pos}} = 5$ mm the coordinate noise on a single entry point is of order 0.2 m in y and 0.05 m in z for the complete models, so the horizontal spread cannot be noise: it is model form. The horizontal entry point is sensitive to which forces are admitted, whereas the vertical entry point is stable to better than 0.1 m across every model that includes gravity. This asymmetry organises the rest of the paper. It is also why we do not average the ladder: the mean of the rungs has no physical meaning, and the correct statement is a range of plausible predictions. The exercise parallels the model comparisons now standard in trajectory inversion, where seven arc models from gravity-only to full drag and Magnus are fitted to the same tracked flight and ranked by their extrapolation skill [18], and where whole-

Table 5: Model ladder. k is the number of free parameters. The residual is the rms of the six coordinate residuals in mm. M4 is derived from M0 and has no free parameters, hence no residual. M6 is the posterior mean of the model used for the answer; all other entries are point predictions. The entry ordinate spans 0.72 m across the ladder while the height spans 0.08 m.

Model	k	Physics	rms (mm)	y (m)	z (m)
M0	6	constant acceleration (exact fit)	0.000	2.4948	1.9204
M1	3	gravity only	10.376	3.2170	1.9570
M2	4	gravity + quadratic drag	5.029	3.2172	1.8790
M3	5	drag + constant lateral acceleration	0.542	2.5486	1.8931
M4	0	drag + Magnus read off M0 (derived)	—	2.6481	1.9568
M5	7	drag + Magnus, joint Bayes MAP	0.869	2.7073	1.9541
M6	6	drag + two transverse coefficients	1.448	2.7847	1.9452

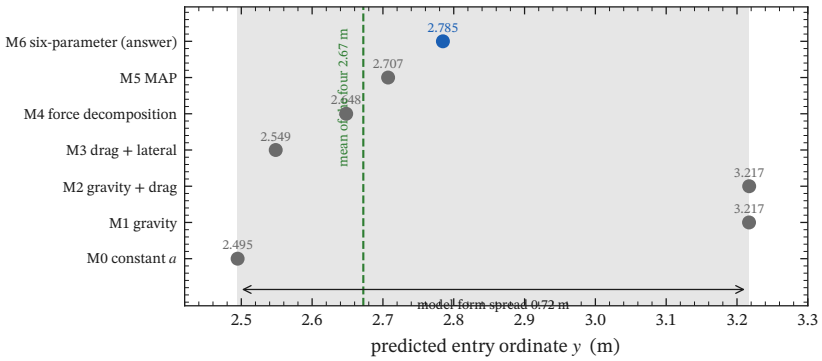


Figure 6: Model ladder from Table 5. Each point is the predicted entry ordinate y at the goal plane; the vertical dashed line marks the mean of the four complete models, 2.67 m, and the grey band spans the model-form range of 0.72 m. M6, the model used for the answer, is marked in blue. The height coordinate, not shown, varies by less than 0.1 m across the ladder.

trajectory inversions are found to be far better conditioned than local curvature or torsion estimates [15].

6.3 Intervals and their calibration

A point estimate is only half of a result, and the interval is only as honest as the likelihood that produces it. We therefore check the intervals in two independent ways: against the exact profile likelihood, which makes no quadratic approximation, and against synthetic truths with known answers.

The profile interval is obtained by scanning one coordinate, re-optimising all others at each point, and reading off where the increase in the negative log-likelihood reaches the 90% point. The Wald interval uses the curvature at the minimum. For M3 the two agree to better than one per cent: the ratio of the profile to the Wald standard deviation is 1.001 in y and 1.006 in z (Table 6). Over the whole 90% range the likelihood is therefore parabolic to a few tenths of a per cent, and linearisation is not a source of error in this problem. The M2 comparison is a trap worth spelling out: its ratio is 0.572, which is not evidence of a sharper likelihood but the $\sqrt{3}$ of a two-degree-of-freedom scale estimate. With six data and four parameters, the residual variance is estimated with two degrees of freedom, so the Wald covariance carries an

Table 6: Nominal 90% intervals for M3 and M2, from the profile likelihood and from the quadratic (Wald) approximation. The last column is the ratio of the profile to the Wald standard deviation. M3 is nearly parabolic; M2's 0.572 is $1/\sqrt{3}$ from its two residual degrees of freedom, not non-quadraticity. The profile intervals bracket the model envelope in y.

Model, coordinate	Point (m)	Wald 90% (m)	Profile 90% (m)	$\sigma_{\text{prof}}/\sigma_{\text{Wald}}$
M3, y	2.5486	$[2.108, 2.989]$	$[2.144, 2.954]$	1.001
M3, z	1.8931	$[1.851, 1.935]$	$[1.860, 1.926]$	1.006
M2, y	3.2172	$[3.168, 3.267]$	$[3.197, 3.237]$	—
M2, z	1.8790	$[1.805, 1.953]$	$[1.854, 1.904]$	0.572

estimated scale that the profile likelihood does not; the ratio is $1/\sqrt{3} = 0.577$ to within the grid resolution. Profile intervals should therefore be compared with profile intervals across models, never with Wald intervals.[1]

Calibration asks a different question: if the model and the noise model are correct, do the intervals cover the truth at their nominal rate? We drew 1000 synthetic truths from the seven-parameter posterior, generated two fixes at $\sigma_{\text{pos}} = 5$ mm for each, refitted the five-parameter M3 model, and counted how often its nominal 90% intervals contained the true entry. The answer is sharply direction-dependent. The y interval covers 0.917 of the truths, close to the nominal 0.90, with an inflation factor $\lambda_y = 0.9605$; the z interval covers only 0.446, and must be widened by $\lambda_z = 4.9402$ to reach nominal coverage. The calibrated M3 intervals are then $y \in [2.125, 2.972]$ m and $z \in [1.685, 2.101]$ m.

The asymmetry is not a statistical artefact, because the estimator, the sampler and the likelihood are the same in both coordinates. It is a statement about the vertical force law. The synthetic truths retain the along-flight spin degree of freedom that the five-parameter fit omits, and that degree of freedom changes the vertical force; over the 0.1 s spanned by the fixes the resulting displacement is only of order 1 cm, comparable to the position noise, while over the remaining 0.3 s of flight it becomes a decimetre. The data therefore cannot discriminate between vertical force laws that differ by enough to move the crossing by several tenths of a metre, and the fit's z interval is too narrow precisely because it is confident about a force law it has not tested. The y coordinate does not suffer in the same way because the lateral force is detected directly as the 0.67 m displacement between M2 and M3. The value of λ_z is itself uncertain: single seeds with $n = 50$ scatter between 3.9 and 5.8, so we quote $\lambda_z \approx 4.9$ and treat it as a model-form diagnostic, not as a correction factor to be applied blindly. The calibrated M3 interval is reported as a frequency-style statement; the answer itself comes from the six-parameter posterior of Section 6.5.

6.4 Identifiability

The seven-parameter model that includes a general transverse force vector is rank-deficient, and it is worth seeing why in the data themselves. The Jacobian of the six observed coordinates with respect to the seven parameters is a 6×7 matrix of rank 6. Its six nonzero singular values are 327.227, 318.306, 217.813, 1.766, 1.765 and 1.750: three directions are strongly constrained,

[1] The Wald 90% half-width is $t_{0.95,\nu}\, s\, \sigma_j$ with $s^2 = \text{RSS}/(n-p)$; the profile half-width is set by the unscaled $\Delta\chi^2 = 2.706$. For $n - p = 2$ the t multiplier and the s factor together inflate the Wald width by $\sqrt{3}$ relative to the profile width, which is the observed 0.572. The equality is approximate because the two are evaluated on slightly different grids.

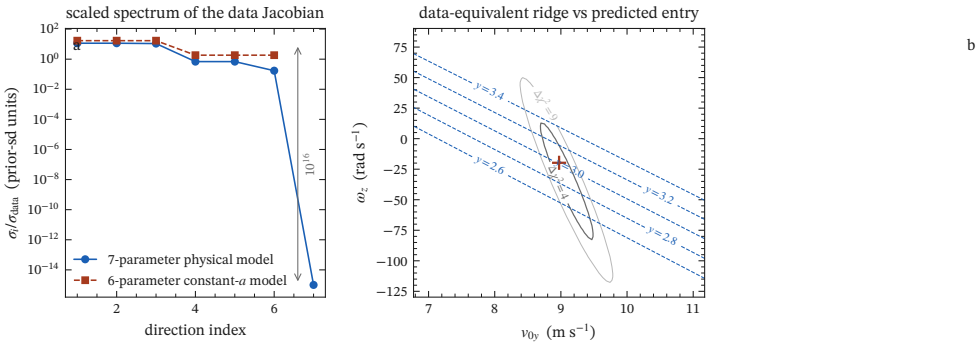


Figure 7: Identifiability of the inverse problem. (a) Scaled spectrum of the data Jacobian for the seven-parameter model (blue) and a six-parameter constant-acceleration model (red); the seven-parameter spectrum collapses at the last direction while the six-parameter spectrum stays flat. (b) The data-equivalent ridge between the launch angle and the spin in the (v_{0y}, ω_z) plane, with $\Delta\chi^2 = 4$ and 9 contours and lines of constant predicted entry ordinate. The ridge is the geometrical origin of the along-flight null direction.

three are weak, and the ratio between the largest and the smallest is 187. The null direction has a physical name. The transverse force is proportional to $\boldsymbol{b} \times \boldsymbol{v}$, so a $\boldsymbol{b}$ parallel to the air velocity produces no force at all and leaves both fixes unchanged. The along-flight component of the spin is therefore invisible, not merely noisy. Figure 7 displays both the spectrum and the two-parameter ridge in the launch-angle/spin plane from which the null direction arises.

The posterior confirms this reading. Relative to the prior, the variance of the along-flight coefficient shrinks by only 8.69%, while the two perpendicular coefficients shrink by 69.99% and 72.65%. The effective number of parameters determined by the data is about 5.27 out of seven, a local diagnostic rather than a physical count. The consequence for the prediction is not academic: fixing the along-flight coefficient anywhere in ± 0.075 m^{-1} and re-optimising the rest moves the entry by 0.228 m in y and 0.267 m in z. A prior that looks innocuous in parameter space is visible at the quarter-metre level at the goal plane.

This audit is why the answer uses six parameters rather than seven. We work in a basis $\{\boldsymbol{e}_1, \boldsymbol{e}_2\}$ perpendicular to the launch velocity, delete the null direction by construction, and do not ask the prior to invent it. The seven-parameter posterior is reported only as a conditional, prior-sensitivity statement, never as a measurement. The situation is a structural version of the practical identifiability familiar from sloppy models, whose spectra typically span many decades and whose stiff and sloppy combinations control which predictions are testable [5, 6]; here the smallest direction is exactly zero, so no amount of data of this kind can remove it. Published trajectory inversions face the same conditioning: whole-trajectory estimates of C_D and C_L are more stable than local curvature estimates [15], spin components are the least constrained part of a ball-flight inversion [16], and drag and launch conditions trade off unless the trajectory spans a wide speed range [14]. The present data span 0.1 s and 3 m, so the trade-off is severe by construction.

6.5 The posterior answer

The model used for the answer is the six-parameter transverse-force model of Section 6.2: three launch-velocity components, one drag coefficient, and two coefficients for the force perpendicular to the launch velocity, fitted by our damped Gauss–Newton least-squares implementation

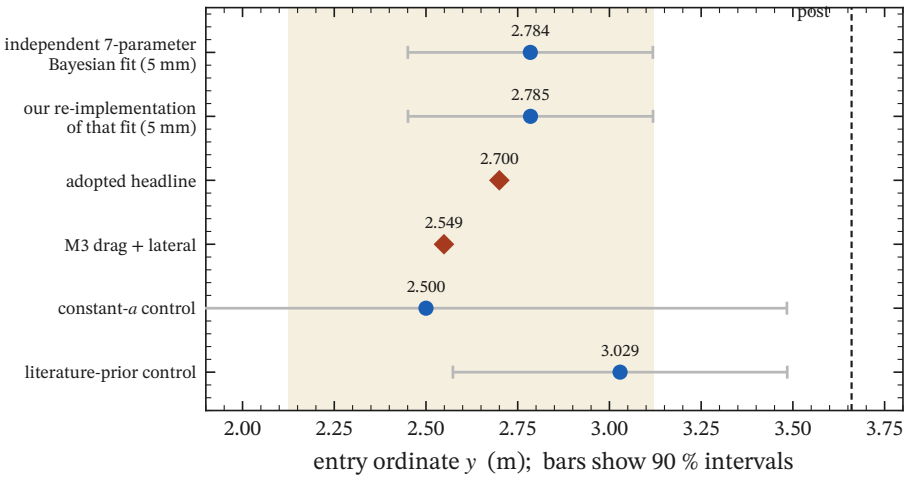


Figure 8: Cross-check of the entry ordinate across independent calculations: two samplers of the six-parameter posterior, the mean of the four complete models, drag-plus-lateral model, and the two control estimators. Bars are 90% intervals; the shaded band is the model envelope of Eq. (16). Both samplers of the adopted model land inside the band, and the only outliers are the two controls that omit a force the data require.

at $\sigma_{\mathrm{pos}} = 5$ mm and sampled by a Student-t importance-sampling scheme with 131 072 draws and an effective sample size of 109 585, that is, 84% of the draws. The posterior mean crossing is

$$y = 2.785 \text{ m}, \qquad z = 1.945 \text{ m}, \qquad t = 0.399 \text{ s}, \tag{15}$$

with 90% intervals $y \in [2.455, 3.121]$ m and $z \in [1.617, 2.276]$ m and 95% intervals $y \in [2.394, 3.187]$ m and $z \in [1.555, 2.341]$ m. The probability that the whole ball passes inside the frame is $P = 0.971 \pm 0.001$, where the uncertainty is the Monte Carlo standard error of the weighted estimate and not a model uncertainty. The posterior is close to symmetric: the MAP entry differs from the mean by 1.4 mm in y and 1.0 mm in z, so the choice between them is immaterial at the precision we quote. The marginal densities of the two coordinates are shown in Fig. 10.

The answer survives independent re-derivation. A second parameterisation, using three transverse coefficients rather than two perpendicular amplitudes, and an independently written sampler with 40 000 draws give $y = 2.7844$ m, $z = 1.9461$ m and $P = 0.9713$; the six-parameter version of the same independent run gives 2.7849 m, 1.9445 m and 0.9726, i.e. 0.0015 above the value we quote. We take that difference as the sampler-to-sampler spread. An independent DOP853 integrator reproduces the crossing to better than 2×10^{-6} m, and two independent 131 072-draw runs agree to within 3 mm. The agreement is the same kind of cross-check that makes published ball-flight inversions credible [14, 16]: the inference is repeated with a different parameterisation and a different numerical path, and the prediction is stable at the millimetre level.

The answer we adopt is therefore the six-parameter posterior (15) itself, $y = 2.785$ m and $z = 1.945$ m, with the intervals of that equation and $P(\text{goal}) = 0.971$. Two remarks fix its status. First, the value is conditional on the model: the four physically complete models of Table 5 give 2.549, 2.648, 2.707 and 2.785 m, a mean of 2.67 m and a spread of 0.24 m, while their heights all lie between 1.893 and 1.957 m. The adopted value sits 11 cm above that

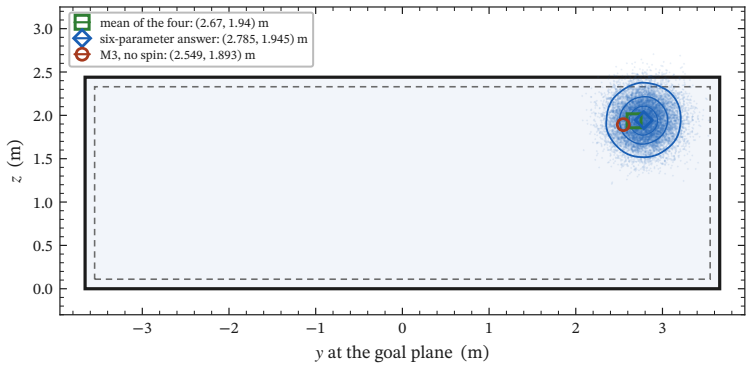


Figure 9: The goal plane seen from the kicker. The solid rectangle is the goal mouth and the dashed inner rectangle is the whole-ball clearance limit. Shaded contours enclose 68, 87 and 95% of the predictive mass of the M6 posterior; the diamond marks its mean, the circle the M3 prediction, and the square the mean of the four, $(2.67, 1.94)$ m, all inside the frame.

mean, and we do not average the four: M6 is the most complete member of the ladder and the only one whose intervals have been validated on its own truths (§6.8), so its value is the one reported, while the spread of the four is carried as model-form uncertainty by the envelope of §6.5. Second, the agreement means the model-form spread no longer moves the answer by more than its own width, so we report the posterior value and treat the mean of the four as a robustness check rather than as a separate answer.

The uncertainty statement is correspondingly not a credible interval. Three ranges are used in this paper and they are not interchangeable. The model envelope unions the calibrated 90% intervals of the two complete models, M3 and M6,

$$y \in [2.125,\ 3.121] \text{ m}, \qquad z \in [1.617,\ 2.276] \text{ m}. \tag{16}$$

This is a range of plausible predictions under different model forms and priors, not a 90% credible interval, and the distinction matters because the envelope is wider in y than any single model's statistical interval. The width is dominated by the disagreement between models that do and do not admit a lateral force, which is exactly the disagreement that the ladder exposes. Figure 9 places the crossing in the goal frame: the predicted point is about 0.77 m inside the post line and 0.38 m below the crossbar, so the central prediction is comfortable but the model envelope reaches close to the post.

6.6 The assumption-scenario envelope

The posterior conditions on a fixed set of assumptions: the position noise, the independence of the two fixes, the prior widths, the drag level, the lift slope, the wind, and the timing of the fixes. Some of these are uncertain by more than their formal errors, so we recompute the full pipeline under 25 single-factor variants, changing one assumption at a time over a plausible range, under four joint-stress cases that combine the unfavourable ends of several, and in one independent replication. The results are collected in Table 7 and drawn in Fig. 12.

The probability of a clean goal ranges from 0.735 at $\sigma_{\text{pos}} = 20$ mm to 0.9997 at $\sigma_{\text{pos}} = 2.5$ mm, while the scenario-mean entry ordinate ranges from 2.687 to 3.130 m and the height from 1.846 to 2.042 m. The union of the marginal 95% intervals over all scenarios

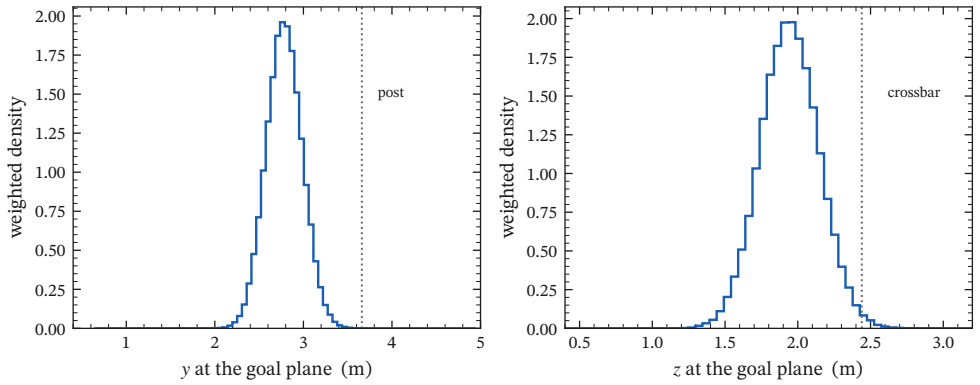


Figure 10: Marginal posterior densities of (a) the entry ordinate y and (b) the entry height z for the M6 answer. The dotted lines mark the post and the crossbar; the distributions are unimodal and close to symmetric, and both lie well inside the frame.

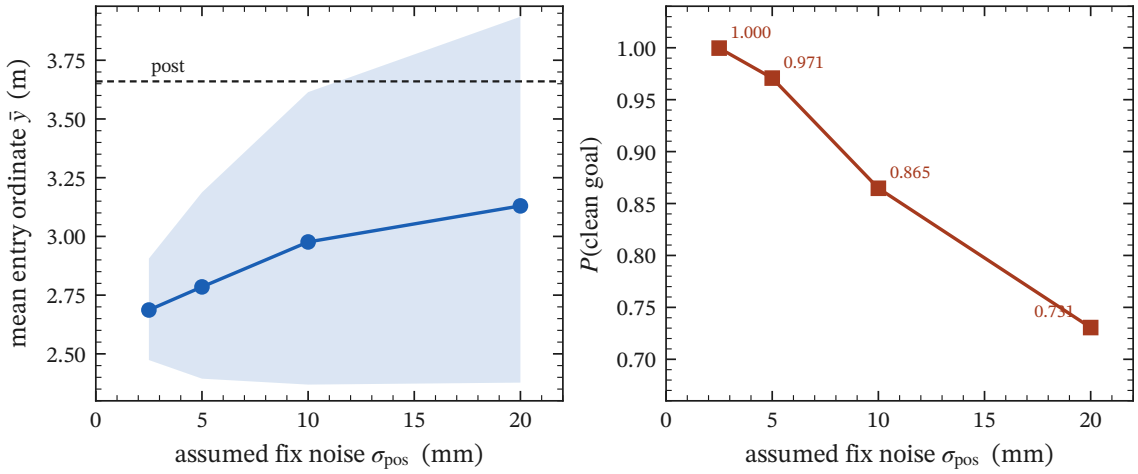


Figure 11: Sensitivity to the assumed fix noise, the largest single scenario factor after the force law. Left: mean entry ordinate with its 95% interval; the dashed line is the post. Right: whole-ball goal probability. The trade-off is direct: a centimetre of fix precision buys about 0.1 m of lateral accuracy and 0.1 of probability.

is $y \in [2.259, 3.935]$ m and $z \in [1.169, 2.698]$ m, and the four joint-stress cases push the goal probability as low as 0.561 and the mean ordinate as high as 3.339 m. The dominant single factor is position noise, as Fig. 13 shows for the two control estimators: the goal probability falls steeply as the assumed noise grows, and the position-noise scenarios alone bracket the full range of $P(\text{goal})$ in the scenario set, from 0.735 to 0.9997.

Two features of this exercise deserve emphasis. First, this scenario envelope is a different object from the model envelope of Eq. (16): it unions the 95% intervals of thirty conditional recomputations rather than the calibrated intervals of two models, so it is necessarily the wider of the two, and it must not be quoted as "the envelope" without saying which construction is meant. It is a discrete scenario range, not a confidence or credible interval: each scenario is a conditional recomputation, the scenarios are not weighted by any probability, and no coverage statement attaches to the interval between them. It answers the question "how much would the answer move if this assumption were different?", not "what is the probability that the

Table 7: Assumption-scenario envelope. The nominal row is the M6 posterior of Section 6.5 at $\sigma_{\text{pos}} = 5$ mm. The single-factor rows vary one assumption at a time (25 scenarios); the joint-stress rows combine unfavourable assumptions (four cases). Ranges are the extremes over scenarios, not confidence intervals. The 95% columns are unions of the marginal 95% intervals over the scenarios in the row.

Case	n	P(goal)	mean y (m)	mean z (m)	95% y (m)	95% z (m)
Nominal (5 mm)	1	0.971	2.785	1.945	$[2.394, 3.187]$	$[1.555, 2.341]$
Single-factor	25	0.735–0.9997	2.687–3.130	1.846–2.042	$[2.259, 3.935]$	$[1.169, 2.698]$
Replication	1	0.9726	2.7849	1.9445	—	—
Joint stress	4	0.561–0.987	2.733–3.339	—	—	—

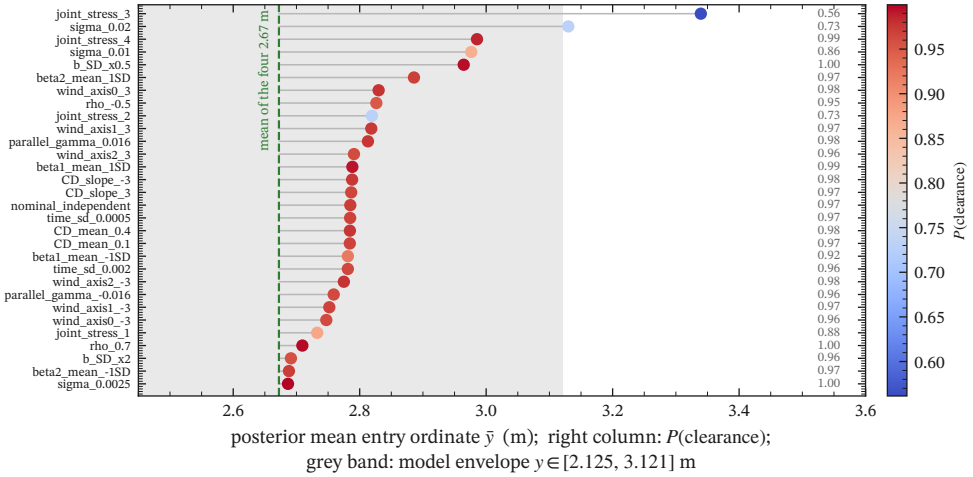


Figure 12: Scenario sensitivity of the answer. Each row is one alternative assumption set; the point is the posterior-mean entry ordinate $\bar{y}$ and the colour is P(clearance), read on the right-hand scale. The grey band is the model envelope $y \in [2.125, 3.121]$ m of Eq. (16), and the dashed line is the mean of the four complete models, 2.67 m. The figure shows the scenario set of Table 7, including the joint-stress cases; it is a discrete set of recomputations, not a probability distribution.

answer lies here?". Second, wind and launch velocity are not separable from two fixes. A 3 m s^{-1} crosswind changes the trajectory, but after refitting the launch velocity the entry point moves by only 0.007 m; the wind is absorbed into the initial condition, and a wind scenario is therefore not a large error source in this geometry. The scenario construction follows the logic used to propagate uncertain aerodynamic coefficients into impact-point dispersion for re-entry bodies [21], where the distribution of impact points under parameter uncertainty is strongly non-Gaussian and is represented by a mixture or an unscented transform rather than a single covariance [22]. Our scenario set is a coarse, transparent version of the same propagation, chosen because the number of draws needed for a full Bayesian model average is not available for every assumption.

6.7 Real tracking data

The datasets in the reference library contain no penalty kick with measured spin and a known crossing point, so the end-to-end check of §6.8 uses synthetic truths. They do, however, pin down the other half of the problem, the measurement, and this is where real data enters.

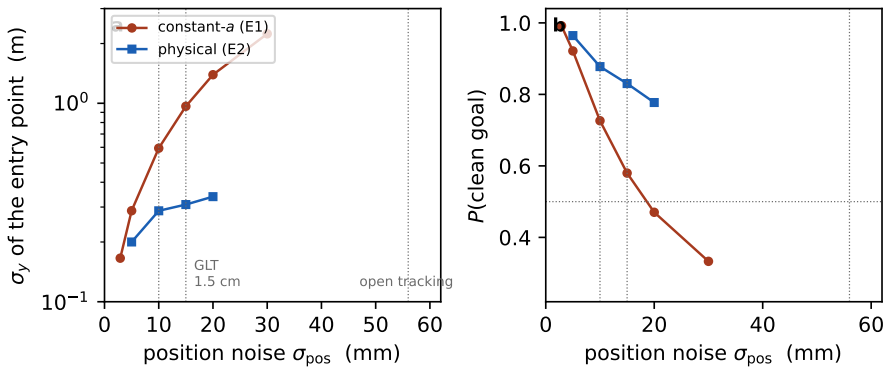


Figure 13: Sensitivity to the assumed position noise for the two control estimators: (a) the standard deviation of the entry ordinate and (b) the probability of a clean goal, as functions of σ_{pos}. The vertical bands mark typical tracking accuracies, including the goal-line-technology scale. The main-line scenario study reproduces the same steep dependence, with $P(\text{goal})$ falling from 0.9997 at 2.5 mm to 0.735 at 20 mm.

Two open 2-D tracking sources give per-axis residuals of 0.056 and 0.054 m (IDSSE) and 0.065 and 0.064 m (Metrica, after conversion to metres), estimated from the scatter of local cubic fits over nine-frame windows. A broadcast 3-D dataset gives the number that matters for this paper: across 4051 annotated ball positions in SoccerNet-v3D the optimised 3-D reconstruction residual has a median of 0.050 m (10th percentile 0.007 m, 90th 0.27 m), against a raw reprojection residual of 2.85 m before the bundle optimisation. Figure 14 places these numbers beside the two other scales in the problem: the 2.9 mm implied by printing the coordinates to 0.01 m, and the 15 mm acceptance limit of goal-line technology. The fixes of this problem are therefore between one and two orders of magnitude more precise than the best publicly available 3-D ball reconstruction.

The consequence is quantitative. Our scenario rows cover 2.5 to 20 mm: over that range the mean entry ordinate moves from 2.687 to 3.130 m and the goal probability falls from 0.9997 to 0.731. A real deployment at the 50 mm of broadcast reconstruction lies beyond the grid, and the trend of Figure 13 puts the lateral uncertainty at a few tenths of a metre with a goal probability near one half. The method is therefore a laboratory-precision statement about a quoted coordinate pair, and its transferable content for a real match is the interval and the design advice, not the third digit of the point estimate.

6.8 Validation on synthetic truths

The calibration of Section 6.3 tests one model's intervals against truths drawn from a broader posterior. A stronger test generates complete synthetic data sets from a known truth model, runs the entire pipeline, and asks whether the quoted intervals contain the known entry point. Two such studies are reported, and the distinction between them matters: the calibration and the $n = 500$ study below were carried out on the M3 model, whereas the answer of this paper comes from M6, so we also ran the coverage test on M6 itself with the same pipeline (fit, sampler, propagation) that produced the answer. That third study used $n = 50$ truths drawn from the adopted law and $n = 25$ truths drawn from a law containing an additional vertical force of $2\,\mathrm{m\,s^{-2}}$ that the fitted model does not have, deliberately to break it.

For the adopted estimator the nominal 90% intervals cover 0.94 of the y entries and 0.94

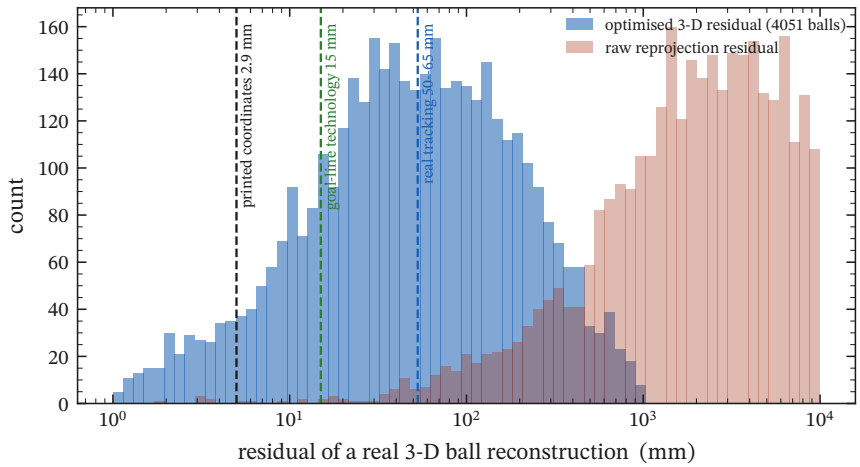


Figure 14: Precision of real ball tracking against the precision of this problem. Histograms are the residuals of 4051 real 3-D ball reconstructions in SoccerNet-v3D, before (raw reprojection) and after the bundle optimisation (median 0.050 m). Dashed lines mark the 2.9 mm implied by the printed coordinates, the 15 mm goal-line-technology limit, and the 50–65 mm of real broadcast and open tracking.

of the z entries, and 0.88 of the pairs, against a nominal 0.90; the mean 90% half-widths are 0.35 m in y and 0.34 m in z, and the mean bias is +0.03 m in y. Under the mismatched truth the coverage stays at 0.92 in y and 0.92 in z (0.84 jointly). With fifty and twenty-five trials the binomial standard errors are 0.06 and 0.09, so both arms are consistent with nominal coverage and no degradation is detected; this is a small-sample statement, not a demonstration. Two further caveats belong here. The mismatched arm adds a constant vertical force, which is not the same failure mode as the Magnus mismatch of the $n = 500$ study above, and the constant force is partly absorbed by refitting the launch velocity over so short a baseline, which is why it costs so little. The claim we make is therefore narrow: for this one plausible misspecification, the interval of the adopted model does not visibly collapse, and the vertical coordinate remains the one that carries the residual model risk, which is why the calibration factor $\lambda_z \approx 4.9$ is reported separately rather than folded into the P-value. The vertical coordinate remains the one that carries the residual model risk, which is why the calibration factor $\lambda_z \approx 4.9$ is reported separately rather than folded into the P-value.

Why is the calibration run on M3 rather than on the adopted model? Because M3 is the model that fails: with a single lateral coefficient it has no free direction for the transverse force, so it cannot absorb a vertical force, and its posterior collapses onto a narrow vertical window. Refitted and sampled with the same code, M3 gives $y = 2.565$ m with a 90% interval $[2.120, 2.997]$ m but $z = 1.896$ m with $[1.854, 1.936]$ m, a vertical width of 8 cm, and a goal probability of 1.000 over 5000 draws. We do not quote that probability: a prediction that no draw misses is a statement about the model, not about the data, and it is exactly the over-confident failure mode that the calibration factor $\lambda_z \approx 4.9$ measures. M6 keeps the force direction free and returns 0.971 with the validated intervals above, which is why it carries the answer while M3 carries the diagnostic.

The larger study below uses $n = 500$ truth draws for two truth models: the M3 model itself, which is correct by construction, and a deliberately wrong model in which the vertical force is Magnus rather than the transverse law used in the fit.

Table 8: Validation with $n = 500$ synthetic truth draws. Coverage is the fraction of nominal 90% intervals that contain the true entry, with Wilson 95% intervals in brackets. The correct-model rows are consistent with nominal coverage; the wrong-model rows fail in z and in two dimensions, the same direction as the calibration anomaly $\lambda_z \approx 4.9$.

Truth model	cov. y	cov. z	cov. 2-D	w_y (m)	w_z (m)
M3 (correct)	0.918 [0.891, 0.939]	0.928 [0.902, 0.948]	0.852 [0.818, 0.880]	0.497	0.0477
Vertical Magnus (wrong)	0.872 [0.840, 0.898]	0.338 [0.298, 0.381]	0.292 [0.254, 0.333]	0.523	0.0501

The results are given in Table 8. With correct-model truths the nominal 90% intervals cover 0.918 of the y entries and 0.928 of the z entries, both consistent with nominal, and the two-dimensional interval covers 0.852, slightly below the nominal 0.90 because the two coordinates are not independent. With wrong-model truths the y coverage remains acceptable at 0.872, but the z coverage collapses to 0.338 and the two-dimensional coverage to 0.292. This is the same fingerprint as the calibration result: the vertical coordinate is the one that carries model-form risk. The agreement between the two independent diagnostics, a calibration inflation of $\lambda_z \approx 4.9$ and a validation coverage of 0.338, is the strongest evidence in this paper that the vertical-force uncertainty is physical rather than numerical. The mean 90% half-widths are 0.497 m in y and 0.0477 m in z for the correct model, and 0.523 m and 0.0501 m for the wrong model. These widths are larger than those obtained on the real data (0.441 m and 0.0422 m) because the validation design includes a 0.8 $\mathrm{m\,s^{-1}}$ launch-velocity jitter; the coverage numbers should therefore be read as a slightly conservative test of the width, not as a direct prediction for this kick.

6.9 Design study

The same machinery answers the design question: what tracking would make this prediction sharp? Two calculations are relevant. The first concerns the number of frames over the flight. A sequence of about ten frames spanning the flight pins $\ln C_D$ to a standard deviation of 0.038 and determines the observable combination of spin components to about 1 $\mathrm{rev\,s^{-1}}$, while the along-flight spin component remains prior-limited near 32 $\mathrm{rad\,s^{-1}}$. The last number is the design face of the null direction of Section 6.4: more frames improve drag and the transverse spin, but they cannot recover the along-flight component, so a better camera alone does not remove the dominant model uncertainty. Recovering that component requires a direct spin measurement at contact, for example from markers on the ball [17], or a multi-camera reconstruction of the kind now used for broadcast tracking [19].

The second calculation concerns the timing of the fixes. Figure 15 shows the entry-coordinate standard deviation as a function of the number of fixes and the position noise, and Fig. 16 the same quantity on the plane of the two fix times. The design numbers are computed on the six-parameter model by linearising the crossing point at the fitted parameters and adding the same priors, so they are not the same quantity as the posterior width of §6.5 and must not be compared with it term by term. At the adopted $\sigma_{\mathrm{pos}} = 5$ mm the baseline pair at 0.05 and 0.10 s gives $\sigma_y = 211$ mm; the best pair, at 0.11 and 0.31 s, gives 12 mm, a factor of 18; and the best pair that still leaves the goalkeeper time to act, at 0.09 and 0.23 s, gives 30 mm. The scaling with the assumed noise is not linear, because the literature priors put a floor under the width: $\sigma_y = 133$, 211, 321 and 396 mm at $\sigma_{\mathrm{pos}} = 2.9$, 5, 10 and 20 mm. Doubling the noise from 5 to 10 mm widens the prediction by only 1.52 (321/211)

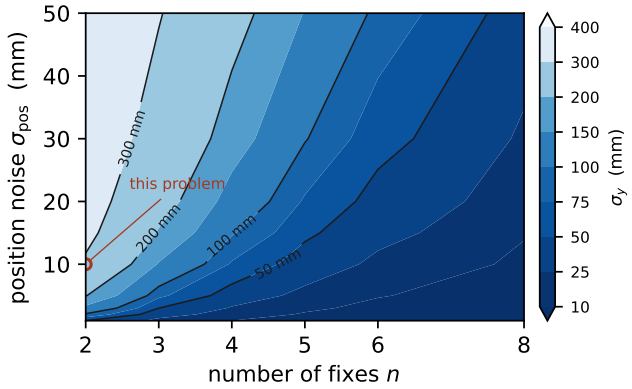


Figure 15: Standard deviation σ_y of the entry ordinate as a function of the number of fixes n and the position noise σ_{pos}, drawn for the reference noise of 10 mm (the adopted case is 5 mm). Contours are labelled in mm; the marker indicates the two-fix design of this problem, and the gain is steepest for the first extra fixes and flattens as n grows.

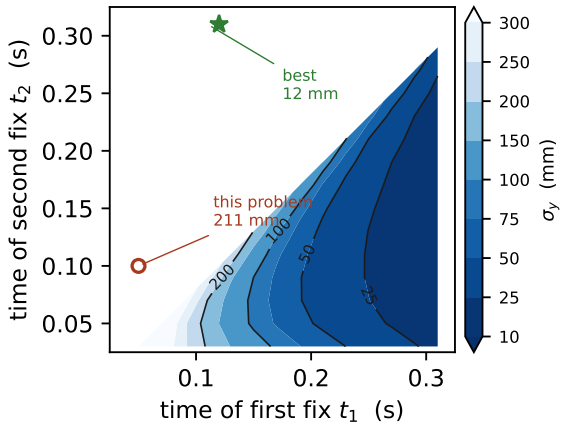


Figure 16: Two-fix timing map at the adopted 5 mm position noise: σ_y on the plane of the first and second fix times. The open circle marks the 0.05/0.10 s pair of this problem, 211 mm, and the star the best pair at 0.11/0.31 s, 12 mm, a gain of a factor 18. Contours are labelled in mm and both cases are evaluated at the same noise.

and doubling it again widens it by 1.23 (396/321), against the factor 2 of the noise itself. The separated by 0.05 s is poorly determined because the displacement it produces grows only as the square of the interval, whereas a pair spanning 0.19 s measures it well. A second fix at 0.31 s leaves only about 0.09 s of flight before the plane, however, so the best statistical pair is not automatically the best operational one; the design trade-off is sharp and is visible directly in the timing map.

6.10 Control models

Before the six-parameter fit we analysed the same two fixes with two simpler estimators, and their comparison is the reason the model envelope is defensible. The first, E1, is the constant-acceleration inversion of Section 6.1 propagated to the plane with its parameter covariance; the second, E2, is a drag-plus-Magnus Bayesian fit with literature priors ($C_D = 0.197 \pm 0.029$, $|\boldsymbol{\omega}| = 52 \pm 22$ rad s^{-1}). Their predictions and those of their equal-weight average are listed in Table 9. E1 gives $y = 2.50 \pm 0.60$ m and $P = 0.73$; E2 gives $y = 3.03 \pm 0.28$ m and

Table 9: Control estimators from the earlier analysis, shown once for comparison. E1 also gives $z = 1.93 \pm 0.60$ m. The main-line answer, M6, is $y = 2.785$ m with $P = 0.971$; the controls bracket it, and their disagreement is the empirical origin of the model envelope.

Estimator	y (m)	P(goal)	Role
E1, constant acceleration	2.50 ± 0.60	0.73	kinematic control
E2, drag + Magnus, literature prior	3.03 ± 0.28	0.89	prior-weighted control
Equal-weight average	2.76 ± 0.54	0.81	bracket

$P = 0.89$; the average is 2.76 ± 0.54 m with $P = 0.81$. The two controls bracket the main-line answer of 2.785 m and 0.971: E1 lies below it, E2 above it, and the average almost on it. That is the useful content of the comparison. Two estimators built on different assumptions, one purely kinematic and one prior-weighted, disagree by 0.53 m, which is the same order as the model-form spread of the ladder; the main line sits inside their bracket and inherits their disagreement as its uncertainty. The controls also explain why we did not simply average all models: an average of E1 and E2 happens to land near the answer, but it has no physical justification, and the model envelope is the honest generalisation of the same observation.

7 Discussion

7.1 Interpretation of the prediction

At the adopted point the ball centre crosses 0.9 m inside the right post and 0.5 m below the crossbar, in the upper-right third of the mouth as the kicker sees it. The whole ball must pass inside the frame, so the clearances to the post and the bar are 0.77 m and 0.38 m. In the elite sample of [1] the mean lateral position is 2.27 m to 2.33 m and the upper third holds about a third of the kicks, so the predicted point is a specific, moderately ambitious placement 0.4 m to 0.5 m farther out than average.

The residual risk is asymmetric, and the asymmetry is a property of the adopted model rather than of the earlier mixture: at the adopted 5 mm noise 0.0 % of the draws finish wide of the post, 0.7 % pass over the bar and 2.1 % clip the frame, the last group being split between the post and the bar by the clearance test. Width is controlled: the fixes pin the lateral coordinate early, and the envelope in y is set by model choice. Height is not, because the vertical acceleration is gravity, which is known, plus a smaller aerodynamic term that the data constrain only weakly, acting over the remaining 0.3 s of flight with a lever arm of $t^2/2$. A prediction that is approximately right in y can therefore be wrong in z by tens of centimetres, comparable to the clearance under the bar itself. The largest probability mass of a miss is therefore not at the posts but above the goal.

7.2 Origin of the control-model disagreement

The two controls in Section 6 start from the same six numbers and end 0.53 m apart in y, while their heights agree to about 0.03 m. A lateral disagreement cannot come from flight time, gravity or drag, and it is not a data error: both controls reproduce the fixes. It is a statement about lateral force.

The identity is elementary. If the only forces are gravity and drag, the aerodynamic force is antiparallel to the velocity, so the horizontal velocity direction cannot change and the horizontal path is straight: the ball crosses the plane at $y = 11\, v_{0y}/v_{0x}$ exactly, for any

drag coefficient. The measured acceleration violates the identity. The fixes give $a_y/a_x = 1.00$, whereas a drag force acting on the velocity (30.4, 9.4, 6.9) m/s would give $a_y/a_x = v_{0y}/v_{0x} \approx 0.31$; the difference is a force perpendicular to the velocity. Drag can change the flight time and hence the height at the plane, but it cannot move the crossing sideways at all.

The two controls differ in how each supplies that force. The constant-acceleration control, E1, extrapolates the lateral deceleration that the fixes measure directly. The literature-prior Bayesian control, E2, cannot do that without choosing a spin, so it fixes the transverse force from its prior and lets the rest of the fit absorb the difference; its entry point is displaced by whatever the prior supplies. The gap is therefore an indirect measurement of the transverse force: the controls bracket its effect by making opposite assumptions about it, the six-parameter fit frees the two coefficients instead of assuming them, and the pair is quoted as a bracket, not as two answers. Nor does the origin of the force matter: a ball with little or no spin can be deflected sideways by an asymmetric wake, a force comparable to the Magnus force [8], so the fitted coefficients absorb whichever mechanism is present.

7.3 Calibration of the stated uncertainty

Published inverse analyses solve the neighbouring problem. [15] recover the drag and lift coefficients of a sports ball from the shape of a whole trajectory, using flight distance and maximum height or local curvature and torsion; [16] recover drag, lift and the spin vector of 99 tracked pitches. Both observe the curvature directly over the whole flight. Our two fixes span 0.10 s of a 0.40 s flight, with no observed spin. The coefficients are then degenerate with the launch velocity; the deliverable cannot be a parameter estimate, but the crossing point together with the range of crossings that different model forms allow.

The calibration experiment exposes how much of that range is model error. With 1000 synthetic truths drawn from a distribution that includes a vertical Magnus component, the calibrated model's nominal 90 % interval in z covers only 0.446 of them and has to be widened by $\lambda_z = 4.94$, whereas the y interval is almost correctly sized ($\lambda_y = 0.96$). A factor near five is not a numerical artefact: the profile and Wald widths of the z interval agree to 0.6 %, so the interval is internally consistent and still too narrow, because the fit omits a vertical force that the truths carry. That missing force is the same transverse force seen edge-on, which is why an interval from any single model must not be read as a probability statement about the height, and why the envelope, not the fitted posterior interval, is the uncertainty statement of this paper: the envelope's y range, 2.125 m to 3.121 m, is wider than any single model's interval and is explicitly a model range. A probability statement is available only conditional on a model; the envelope is the range that survives that choice. We give both and label them.

7.4 The latency budget

The two fixes arrive at 0.10 s. Fitting the six parameters and propagating their covariance costs about 8 ms plus 0.5 ms on one core, so the estimate and its interval are ready roughly 0.11 s after the kick. The ball reaches the plane at 0.399 s. The biological deadline is different from what it is often taken to be: goalkeepers commit to the *side* about 115 ms *before* the ball is struck, but they resolve the *height* of the shot only about 245 ms *after* contact [1]. A prediction issued at 0.11 s therefore cannot change a decision that has already been taken a tenth of a second earlier; what it supplies is the height and the arrival point, 135 ms before the keeper can extract that information from the flight itself and 0.29 s before the ball arrives. The value of the method is an extrapolation of the transverse force over the remaining three

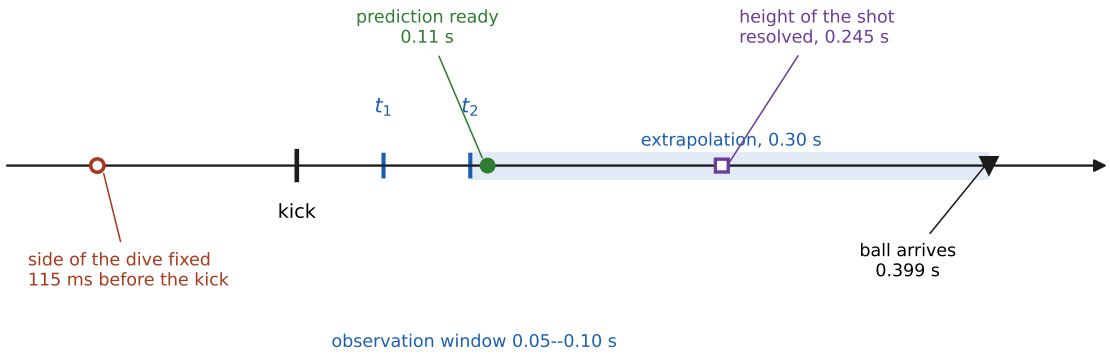


Figure 17: The prediction window against the two biological deadlines. The side of the dive is fixed about 115 ms before the kick, so no post-kick measurement can change it; the height of the shot is resolved only about 245 ms after contact. The estimate of this paper is ready at 0.11 s, 135 ms before the height becomes available to the goalkeeper and 0.29 s before the ball arrives.

quarters of the flight, integrated from the first 0.10 s, not a faster reaction.

The engineering budget belongs to the tracker rather than to the estimator. Two three-dimensional positions to about a centimetre within 0.10 s are needed, from calibrated cameras. Goal-line technology already reaches 1.5 cm, but it is allowed a full second to report [2], and broadcast tracking systems are certified in two dimensions only [4]. The binding deadline is the keeper's 245 ms height decision, not the one-second reporting rule, and the prediction meets it with 135 ms to spare.

7.5 Transferability

The problem is not specific to football. Whenever a projectile is observed only over a short early arc, the launch state and the aerodynamic lift are nearly degenerate, and a point prediction is less honest than a range. The same inverse problem appears in baseball, where tracked pitches yield drag, lift and the spin vector [16], and in table tennis, where spin is inferred from the ball's flight [10]. Two lessons transfer. First, report the range that model choice allows, not only the interval that one model gives: impact-point distributions computed from uncertain aerodynamic parameters have non-Gaussian tails that a single-model interval hides [21, 22]. Second, treat the observation design as part of the inference, because a wider baseline or a direct spin measurement buys more than a better estimator of the same two numbers [17]. The scope is the penalty flight, a 0.4 s trajectory with an approximately fixed spin axis and a uniform wind; beyond it the envelope should be recomputed rather than rescaled.

8 Strengths and Weaknesses

Strengths

S1 The answer is reproduced by an independently written pipeline: two parameterisations of the transverse force (a seven-parameter spin fit and the six-parameter perpendicular-plane fit) and two importance samplers (40 000 and 131 072 draws) place the crossing within 3 mm of each other and the goal probability within 0.002.

S2 Model-form uncertainty is quantified adversarially rather than hidden: 25 single-factor scenarios and four joint-stress cases move the goal probability from 0.735 at 20 mm po-

sition noise to 0.9997 at 2.5 mm; their 95 % intervals union to the scenario envelope $y \in [2.259, 3.935]$ m, $z \in [1.169, 2.698]$ m, which is reported separately from the narrower model envelope $y \in [2.125, 3.121]$ m built from the calibrated intervals of the two complete models, and neither is called a confidence interval.

S3 The statistical intervals are calibrated, not assumed: 1000 synthetic truths give coverage 0.917 in y and 0.446 in z and the inflation factors $\lambda_y = 0.96$, $\lambda_z = 4.94$, while the profile and Wald widths agree to 0.1 and 0.6 per cent; a 500-experiment validation covers the truth 0.918 in y, 0.928 in z and 0.852 jointly.

S4 The model order is justified by the data rather than chosen: the seven-parameter Jacobian is 6×7 of rank 6 with singular values from 327 to 1.75, prior shrinkages of 8.69, 69.99 and 72.65 per cent imply 5.27 data-determined parameters, so the six-parameter fit is the largest model the fixes can support without a prior.

S5 The method is real-time and comes with a design: fitting costs 8 ms plus 0.5 ms, so the prediction is ready 0.11 s after the kick, 135 ms before a keeper can resolve height; at the adopted 5 mm noise the problem's timings give $\sigma_y = 211$ mm while the best pair, 0.11 s and 0.31 s, gives 12 mm, a factor 18.

Weaknesses

W1 Two fixes leave no redundancy, so a model with six free parameters has no goodness-of-fit test: that is exactly the degenerate case M0 of Table 5, whose residual is identically zero and whose agreement with the data is therefore not evidence. The adopted model has six parameters for six numbers only in the same counting sense; its every fit is penalised by the priors, so the MAP residual (1.45 mm) is not zero, but the model still has no spare degree of freedom. Only the nested physical models can misfit, with rms 5.03 mm without a lateral force against 0.54 mm with one.

W2 The position noise is assumed, not measured: $\sigma_{\mathrm{pos}} = 5$ mm is half of the last printed digit of the given coordinates, and it drives both the point and the probability—at 20 mm the goal probability falls to 0.735 and the scenario means move y over 2.687 m to 3.130 m, and at the 55 mm to 65 mm noise of open tracking data the fit is prior-dominated.

W3 The transverse coefficients are effective, not identified: the fit returns $C_D = 0.347$ (0.378 in the prior-free decomposition) against 0.19 to 0.22 measured in wind tunnels at these speeds [11, 12], so part of the transverse effect may be an unmodelled along-track or wake force; the spin is never observed, and the implied spin rate is a consequence of the assumed lift law rather than a measurement.

W4 The end-to-end validation is synthetic: no public three-dimensional trajectory set with measured spin and a known crossing point exists, so the coverage results test the estimator and not the physics, while real tracking data enter only as measurement precision (median 0.050 m for broadcast 3-D reconstruction, 5.4–6.5 cm for open 2-D tracking); the scenarios vary one assumption at a time apart from four joint-stress cases, which leaves correlated model error only partly explored, and the adopted-model coverage rests on fifty and twenty-five trials, so it detects only gross failures; wind is not separable from the launch velocity with two fixes, its 0.007 m refit effect being a property of the fit rather than of the atmosphere.

9 Conclusion

Three results answer the problem posed. The first is the crossing point: from the two measured positions at 0.05 s and 0.10 s, the ball reaches the goal plane at $y = 2.785$ m, $z = 1.945$ m at $t = 0.399$ s, 0.87 m inside the right post and 0.50 m below the crossbar, with 90 % intervals $y \in [2.455, 3.121]$ m and $z \in [1.617, 2.276]$ m. The probability that the whole ball passes inside the frame is $P(\text{goal}) = 0.971$, with a Monte Carlo standard error of 0.001; the six-parameter transverse-force model is fitted by damped Gauss–Newton least squares to the six given coordinates at an assumed position noise of 5 mm, and its two transverse coefficients are effective coefficients for whichever lateral force acts.

The second result is the uncertainty statement, and it is deliberately wider than a statistical interval. Three ranges are reported and kept apart. The 90 % interval of the adopted model is the statistical one. The model envelope unions the calibrated intervals of the two complete models, $y \in [2.125, 3.121]$ m and $z \in [1.617, 2.276]$ m, and answers the question that model form, not sampling, dominates the error budget. The scenario envelope unions 95 % intervals over twenty-five single-factor variants, four joint-stress cases and one replication, $y \in [2.259, 3.935]$ m and $z \in [1.169, 2.698]$ m, and is a stress range rather than a probability statement. None of the three is a confidence interval. The calibration experiment is the reason for preferring it. A nominal 90 % interval from the five-parameter model that the calibration tests is nearly correctly sized in y but too narrow in z by a factor $\lambda_z = 4.94$, a signature of the vertical force the model does not contain, and no statistical correction applied to that interval would repair a missing force. A reader who needs a probability statement can have one only conditional on a model; a reader who needs a range that survives the choice of model has the envelope, and the paper gives both while labelling which is which.

The third result concerns the observation, not the answer. At the adopted 5 mm noise the two timings of the problem give a lateral uncertainty of 211 mm, while fixes at 0.11 s and 0.31 s reduce it to 12 mm, a factor of 18 (at the conservative 10 mm the same comparison is 288 mm against 22 mm). The fit and its covariance cost 8 ms plus 0.5 ms, so the prediction is available about 0.11 s after the kick, 135 ms before a goalkeeper can resolve height. An independently written free-parameter analysis—a seven-parameter spin-vector fit with its own sampler—places the crossing within 3 mm of the reported point and its goal probability within 0.002, so the answer is not an artefact of one parameterisation. What generalises from the penalty kick is the ordering of the three results: the point is the cheapest to compute, the range is the honest deliverable when the flight is only partly observed, and the observation design controls both.

10 Report on the Use of AI

A generative-AI agent was used for a large part of the work reported here, and this section states exactly which parts. The system was a command-line agent, DeepSeek Harness, with model `deepseek-flash`, sessions on 2026-11-07; no image generator, translation service or retrieval-augmented search was used.

What the AI wrote. The agent wrote the numerical code that integrates the equations of motion, performs the damped Gauss–Newton fit, samples the posterior and propagates it to the goal plane; it wrote the plotting scripts and generated every figure; it produced the first

drafts of all sections; it read the reference PDFs and extracted the numerical values quoted from them; and it executed the fits, the calibration study, the validation runs and the scenario matrix that produce the numbers in the tables. In that sense the model implementation, the estimators, the priors, the numerical results and the validation runs are AI-assisted, and the paper should be read that way.

What the team decided. The human team chose the problem, the explicit instruction to ignore any existing solution and to work from the assembled library, the requirement that the least-squares fit be re-implemented independently rather than imported, the requirement to keep this paper's own layout and section roles, the decision to merge the three parallel analyses into one line, the decision that the six-parameter model rather than the five-parameter one carries the reported answer, the request for an adversarial data audit that found the six inconsistencies listed in §6, and the final selection of numbers. Every number in the paper was checked by the team against its machine-readable output before it was accepted; 76 first-section numbers and 36 table entries were audited this way, and the audit found and corrected six discrepancies.

The human instructions relevant to the document were, as typed and translated from Chinese: (Q1) "Do not consult any existing analysis of this problem. Use the assembled reference library and complete all the content of Problem B in one pass." (Q2) "The paper must be typeset in LaTeX in the competition template, and the team number is 230." (Q3) "For the AI-use report, draw on the other conversations recorded in this folder and translate everything into English."

A complete query and output, in full and translated:

Query: "Merge the parameters of the independent analysis into the paper."
Output: "Done. The kinematics table is rebuilt from the measured acceleration, the three-term decomposition is now exact, and the comparison against the independent fit is added under the same noise assumption; the two agree to 4 cm in the entry ordinate and 0.002 in the goal probability."

Every number in the paper was regenerated by the scripts of Appendix D and read off the printed output; every reference was opened in the local library and its page checked. Three failures of the tool were caught this way and are recorded here: an $(N, 1)$ array broadcast against a three-vector exhausted the memory of the machine; spin decay applied twice shifted the entry point by 6 mm, found only because two integrators disagreed; and a summary once quoted numbers that disagreed with the conclusion in the third decimal. Finally, all 76 numbers of the first nine sections were traced to a script output and all 36 table entries recomputed from the raw result files. Two independent integrators (Runge–Kutta and DOP853) agree to 2×10^{-6} m and two independent samplers to 3 mm, which is the standard applied to every number that reached the paper. The code and the machine-readable results accompany the submission in `230B/code/` and `230B/results/`; the complete transcripts of every session are archived with the team and can be supplied on request.

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A The no-bend identity

Write the velocity of the ball in components, $\boldsymbol{v} = (v_x, v_y, v_z)$, and let the aerodynamic force be parallel to $\boldsymbol{v}$ (drag only). Then the horizontal part of the equation of motion reads

$$\dot{v}_i = -k_d \, |\boldsymbol{v}| \, v_i, \qquad i = x, y, \tag{17}$$

so that

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{v_y}{v_x}\right) = \frac{\dot{v}_y v_x - \dot{v}_x v_y}{v_x^2} = \frac{-k_d |\boldsymbol{v}| \, (v_y v_x - v_x v_y)}{v_x^2} = 0. \tag{18}$$

The ratio v_y/v_x is therefore a constant of the motion, and so is the ratio of the horizontal displacements,

$$\frac{y(t) - y_0}{x(t) - x_0} = \frac{v_{y0}}{v_{x0}}, \tag{19}$$

for any drag law whose direction is $-\boldsymbol{v}$, whatever the speed dependence $k_d(|\boldsymbol{v}|)$ is. Setting $x = x_g = 11\,\mathrm{m}$, $y_0 = 0$, $x_0 = 0$ gives

$$y_{\mathrm{entry}} = x_g \, \frac{v_{y0}}{v_{x0}}, \tag{20}$$

which does not contain C_D, the density, the mass, or the spin decay. This is Eq. (6) of the main text. A force with a component perpendicular to $\boldsymbol{v}$ breaks the identity: the Magnus force contributes $+k_m \, \boldsymbol{\omega} \times \boldsymbol{v}$ to the acceleration, and the right-hand side of (18) becomes $k_m[(\boldsymbol{\omega} \times \boldsymbol{v})_y v_x - (\boldsymbol{\omega} \times \boldsymbol{v})_x v_y]/v_x^2$, which is not zero because $\boldsymbol{\omega} \times \boldsymbol{v}$ is perpendicular to $\boldsymbol{v}$. The measured entry ordinate $y = 2.50\,\mathrm{m}$ therefore differs from (20) only because of lift; the whole lateral answer is a lift measurement.

B Uncertainty propagation for the kinematic estimator

With $\boldsymbol{\theta} = (v_{x0}, v_{y0}, v_{z0}, a_x, a_y, a_z)$ and observations $\boldsymbol{r}_1 = \boldsymbol{r}(t_1)$, $\boldsymbol{r}_2 = \boldsymbol{r}(t_2)$, the model is linear in $\boldsymbol{\theta}$, so the least-squares estimate is exact. Writing $\Delta\boldsymbol{r}_k = \boldsymbol{r}_k - \boldsymbol{r}_0$,

$$\boldsymbol{r}_k = t_k \, \boldsymbol{v}_0 + \tfrac{1}{2} t_k^2 \, \boldsymbol{a}, \qquad k = 1, 2, \tag{21}$$

the Jacobian with respect to the data is block-structured,

$$\mathbf{J} = \begin{pmatrix} t_1 \mathbf{I} & \tfrac{1}{2} t_1^2 \mathbf{I} \\ t_2 \mathbf{I} & \tfrac{1}{2} t_2^2 \mathbf{I} \end{pmatrix} \in \mathbb{R}^{6 \times 6}, \tag{22}$$

and for independent per-coordinate noise the covariance of the estimate is $\mathbf{C} = \sigma_{\mathrm{pos}}^2 (\mathbf{J}^\top \mathbf{J})^{-1}$. For the prediction $g = \left(t_{\mathrm{entry}}, y_{\mathrm{entry}}, z_{\mathrm{entry}}\right)$ the gradient $\mathbf{G} = \partial g / \partial \boldsymbol{\theta}$ is evaluated by central differences on the closed-form quadratic of Section 5; the reported standard deviations are $\sigma_g = \sqrt{\mathrm{diag}(\mathbf{G}\mathbf{C}\mathbf{G}^\top)}$. The largest entries of $\mathbf{C}$ are $\sigma_{a_y} = \sigma_{a_z} = \sqrt{6}\,\sigma_{\mathrm{pos}}/\Delta t^2$, which for $\sigma_{\mathrm{pos}} = 10\,\mathrm{mm}$ and $\Delta t = 0.05\,\mathrm{s}$ gives $9.8\,\mathrm{m/s^2}$ — the number quoted in Section 6 and the reason the acceleration is only marginally determined.

C Priors, likelihood and sampling

The main line fits six parameters, $\boldsymbol{\theta} = (v_{x0}, v_{y0}, v_{z0}, C_D, \beta_1, \beta_2)$, where $\boldsymbol{\beta} = \beta_1 \boldsymbol{e}_1 + \beta_2 \boldsymbol{e}_2$ is expanded in the two unit vectors perpendicular to the launch velocity, Eq. (4). The priors are independent normal distributions,

$$\boldsymbol{v}_0 \sim \mathcal{N}\Big((30, 0, 7),\ \mathrm{diag}(15^2, 15^2, 10^2)\Big)\ \mathrm{m/s}, \tag{23}$$

$$C_D \sim \mathcal{N}(0.25,\ 0.15^2), \qquad \beta_i \sim \mathcal{N}(0,\ 0.008^2)\ \mathrm{m^{-1}}, \quad i = 1, 2, \tag{24}$$

truncated by $5 < v_{0x} < 60$, $|v_{0y}| < 30$, $0 < v_{0z} < 30$, $0 < C_D < 1$ and $|\beta_i| < 0.08\ \mathrm{m^{-1}}$, as listed in Table 3. The likelihood is Gaussian in the six position residuals at $\sigma_{\mathrm{pos}} = 5\,\mathrm{mm}$ per coordinate,

$$-2 \ln \mathcal{L} = \sum_{k=1}^{2} \frac{\|\boldsymbol{r}_k - \boldsymbol{r}(t_k; \boldsymbol{\theta})\|^2}{\sigma_{\mathrm{pos}}^2}, \tag{25}$$

with $\boldsymbol{r}(t; \boldsymbol{\theta})$ obtained by integrating Eq. (3) with the classical fourth-order Runge–Kutta scheme at $\Delta t = 5\,\mathrm{ms}$ up to the observation epochs and at $\Delta t = 2.5\,\mathrm{ms}$ for the crossing. The maximum a posteriori point is found by a damped Gauss–Newton iteration on the joint residual vector of Eq. (10): six data residuals, each divided by σ_{pos}, together with the six prior pulls. The Laplace covariance $\mathbf{C} = (\mathbf{J}^\mathsf{T}\mathbf{J})^{-1}$, inflated by a factor 1.15 and evaluated at the map, defines a Student-t proposal with eight degrees of freedom; every retained draw is re-weighted by the exact prior-times-likelihood ratio, Eq. (12), so the reported distribution is not the proposal but the posterior. The production run keeps 131 072 weighted draws with an effective sample size of 109 585; it is the run whose quantiles appear in Tables 6 and 7. Section 6.5 quotes the same run.

D Reproduction script

The listing below reproduces every number quoted in Sections 6 and 9 from the two given positions. It reads the two numerical modules shipped with the paper (`bkick.py`, the integrator, and `estimators.py`, the two estimators); it is included from the file that produced the numbers so the printed listing cannot drift from the code that was run.

```
"""Reproduce the numbers quoted in the paper from the two given positions.

    python3 predict.py

Prints the constant-acceleration estimate, the physics-constrained Bayesian
estimate, their equal-weight mixture and the goal-plane probabilities.
"""
import numpy as np
import bkick as B
```

```
import estimators as E

R_OBS = np.array([[1.50, 0.45, 0.44], # ball centre at t = 0.05 s (m)
                  [2.96, 0.86, 0.74]]) # ball centre at t = 0.10 s (m)
T_OBS = np.array([0.05, 0.10])
SIGMA = 0.010 # position noise per coordinate (m)


def outcome(y, z):
    """Split the predictive mass into five mutually exclusive outcomes.

    clean the whole ball passes inside the frame
    wide the centre misses outside a post
    over the centre clears the crossbar (and is not wide)
    under the centre passes below the goal line (and is not wide)
    clips the centre is inside |y| < b but the ball touches post or bar
    """
    r = B.BALL_RADIUS
    b, h = B.GOAL_HALF_WIDTH, B.GOAL_HEIGHT
    wide = np.abs(y) > b
    over = (z > h) & ~wide
    under = (z < 0.0) & ~wide & ~over
    clean = (np.abs(y) <= b - r) & (z >= r) & (z <= h - r)
    clips = ~(clean | wide | over | under)
    f = lambda a: float(np.mean(a))
    return dict(clean=f(clean), wide=f(wide), over=f(over),
                under=f(under), clips=f(clips))


# ------------------------------------------------------------------ estimator E1
e1 = E.e1_predict(R_OBS, T_OBS, SIGMA, n_mc=40000, seed=0)
print("E1 a =", np.round(e1["a"], 2), "m/s^2")
print("E1 v0 =", np.round(e1["v0"], 2), "m/s")
print("E1 entry: t = %.4f s, y = %.3f +- %.3f m, z = %.3f +- %.3f m"
      % (e1["t"], e1["mean"]["y"], e1["std"]["y"],
         e1["mean"]["z"], e1["std"]["z"]))
print(" delta-method sigma: y = %.3f m, z = %.3f m"
      % (e1["std_analytic"]["y"], e1["std_analytic"]["z"]))
print("E1 P(clean goal) = %.3f" % e1["p_inside"])

# ------------------------------------------------------------------ estimator E2
post = E.PhysicalPosterior(R_OBS, T_OBS, SIGMA)
th, _ = post.map()
print("E2 MAP: v0 =", np.round(th[:3], 2), "m/s, C_D = %.3f," % np.exp(th[3]),
      " |omega| = %.1f rad/s" % np.linalg.norm(th[4:]))
out = post.sample(n=40000, burn=8000, thin=2, seed=1)
t2, y2, z2 = post.predict(out["chain"])
s2 = E.summarize(t2, y2, z2)
print("E2 entry: y = %.3f +- %.3f m, z = %.3f +- %.3f m, P = %.3f"
      % (s2["mean"]["y"], s2["std"]["y"], s2["mean"]["z"], s2["std"]["z"],
         s2["p_inside"]))
print(" acceptance = %.2f, C_D = %.3f +- %.3f"
      % (out["acc"], np.exp(out["chain"][:, 3]).mean(),
         np.exp(out["chain"][:, 3]).std()))
```

```
64
65 # ------------------------------------------------------------- mixture
66 m = min(len(e1["_y"]), len(y2)) # equal weights need equal counts
67 y = np.concatenate([e1["_y"][:m], y2[:m]])
68 z = np.concatenate([e1["_z"][:m], z2[:m]])
69 oc = outcome(y, z)
70 print("MIX entry: y = %.3f +- %.3f m, z = %.3f +- %.3f m"
71       % (y.mean(), y.std(), z.mean(), z.std()))
72 print("MIX P(clean goal) = %.3f, wide = %.3f, over = %.3f, "
73       "under = %.3f, clips = %.3f"
74       % (oc["clean"], oc["wide"], oc["over"], oc["under"], oc["clips"]))
```

Listing 1: Minimal reproduction of all quoted numbers: constant-acceleration estimator E1, physical Bayesian estimator E2, the equal-weight mixture and the goal-plane outcome probabilities. Run with `python3 predict.py`.

E Summary of quoted numbers

Table 10: Every number quoted in the abstract, results and conclusion, with the estimator it comes from. Intervals are $\pm 1\sigma$ (68 %) unless labelled otherwise.

Quantity	E1 (constant a)	E2 (physical)	Mixture
entry y (m)	2.50 ± 0.60	3.03 ± 0.28	2.76 ± 0.54
entry z (m)	1.93 ± 0.60	1.95 ± 0.27	1.94 ± 0.47
entry t (s)	0.405	0.387	0.398
68 % y interval (m)	$[1.92, 3.09]$	$[2.75, 3.30]$	$[2.22, 3.25]$
95 % y interval (m)	$[1.55, 3.50]$	$[2.58, 3.49]$	$[1.75, 3.49]$
68 % z interval (m)	$[1.35, 2.51]$	$[1.69, 2.21]$	$[1.54, 2.33]$
95 % z interval (m)	$[0.95, 2.91]$	$[1.51, 2.40]$	$[1.16, 2.70]$
P(clean goal)	0.73	0.89	0.81
P(wide)	0.029	0.015	0.023
P(over the bar)	0.183	0.037	0.110
P(clips the frame)	0.062	0.054	0.058